

FACULTY OF ENGINEERING AND NATURAL SCIENCES

DEPARTMENT OF AUTOMOTIVE ENGINEERING



Design and Thermal Analysis of an Improved Cooling System for a Four-Cylinder

Gasoline Engine

A GRADUATION PROJECT

submitted by

MAJD NA MOFLEH

in partial fulfillment of the requirements for the degree of

BACHELOR OF SCIENCE

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Program: Automotive Engineering

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by

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Submitted to the Department of Automotive Engineering of

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ABSTRACT

Design and Thermal Analysis of an Improved Cooling System for a Four-Cylinder Gasoline Engine.

Poor cooling system performance can lead to reduced efficiency, accelerated component degradation, and potential engine failure. Modern high-performance gasoline engines, in particular, require more advanced thermal management strategies to maintain stable operating temperatures under variable driving conditions. Conventional cooling systems, typically composed of a mechanical water pump, a single-loop coolant circuit, and a standard aluminum radiator often struggle to eliminate hot spots, ensure uniform coolant flow, and efficiently reject heat during high-load operation.

This project proposes the design and thermal analysis of an improved cooling system for a four-cylinder gasoline engine. The design integrates advanced system components, including an electronically controlled dual-loop cooling circuit, an electric water pump capable of variable flow control, and a high-efficiency radiator with optimized microchannel architecture. The proposed solution aims to enhance heat rejection, reduce temperature gradients within the engine block, and improve overall thermal efficiency. Computational tools will be employed to simulate coolant flow behavior, pressure distribution, and heat transfer characteristics across different operating conditions.

The study includes a comparative evaluation between the conventional cooling architecture and the improved system. Performance metrics such as coolant temperature uniformity, heat rejection rate, energy consumption, and response under

Transient conditions will be used to assess the effectiveness of the proposed design. Expected outcomes include enhanced thermal stability, improved component longevity, reduced parasitic losses, and better adaptability to modern engine requirements.

This project aims to provide a feasible, analytically validated cooling solution that can be integrated into small-to-medium displacement gasoline engines, contributing to improved performance, efficiency, and reliability in automotive applications.

Keywords: Cooling System, Thermal Analysis, Four-Cylinder Engine, Heat Transfer, Radiator Design, Electric Water Pump.

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SYMBOLS

No.	Symbol	Description	Unit
1	$\dot{m}$	Mass flow rate	kg/s
2	c_p	Specific heat capacity	J/kg · K
3	$\dot{Q}$	Heat transfer rate	W
4	$\dot{Q}_{\text{head}}$	Heat removed Cylinder head	W
5	$\dot{Q}_{\text{block}}$	Heat removed Engine block	W
6	$\dot{Q}_{\text{total}}$	Total heat transfer rate	W
7	T	Temperature	°C
8	T_{in}	Coolant inlet temperature	°C
9	T_{out}	Coolant outlet temperature	°C
11	ΔT	Temperature difference	°C
12	α	Valve opening ratio	—
14	τ	Thermal response	s
15	$T(t)$	Temperature as a function of time	°C
16	T_{target}	Target operating temperature of the component.	°C
17	$\dot{Q}_{\text{rad}}$	Radiator heat rejection	W
18	T_0	Initial temperature at cold-start conditions	°C

ABBREVIATIONS

No.	ABBREVIATION	DESCRIPTION
1	ICE	INTERNAL COMBUSTION ENGINE
2	EWP	ELECTRIC WATER PUMP
3	ECU	ENGINE CONTROL UNIT
4	CAD	COMPUTER-AIDED DESIGN

I. INTRODUCTION

Section 1 – Overview

1.1 Project Background

Internal combustion engines generate significant thermal loads during operation due to fuel combustion and mechanical friction. If this heat is not effectively removed, excessive temperatures can lead to reduced efficiency, accelerated material degradation, and potential engine failure [5], [12].

Conventional cooling systems used in four-cylinder gasoline engines often exhibit limitations in coolant distribution, heat rejection capacity, and adaptability to varying operating conditions. These limitations may result in non-uniform temperature distribution and the formation of local hot spots, particularly in critical regions such as the cylinder head [1], [13].

Therefore, improving engine thermal management has become essential for enhancing performance, efficiency, and durability.

This project focuses on the design and thermal analysis of an improved cooling system that enhances temperature uniformity, minimizes hot spots, and improves overall engine thermal performance through controlled coolant flow distribution.

1.2 Project Objectives

The main objectives of this project are:

- To design an improved cooling system for a four-cylinder gasoline engine
 - To reduce temperature gradients and eliminate hot spots in critical engine regions
 - To enhance coolant flow distribution and heat rejection efficiency
 - To evaluate system performance using analytical and numerical thermal analysis
 - To compare the proposed system with a conventional cooling system
-

1.3 Project Scope

The scope of the project includes:

- Conceptual design of the engine cooling system architecture.
 - Selection of key cooling components, including the radiator and water pump.
 - Thermal analysis using analytical calculations and numerical methods
 - Performance comparison between different conceptual cooling system designs
 - Documentation and presentation of the design process and results
-

1.4 Alternative Conceptual Solutions

Several conceptual cooling system configurations were considered:

1.4.1 Conventional Cooling System with Improved Radiator

This concept enhances the existing single-loop cooling system by introducing a high efficiency aluminum radiator with improved fin density and airflow management [8],[14].

1.4.2 Electrically Controlled Cooling System

This concept replaces the mechanical water pump with an electric pump and electronically controlled thermostat, allowing variable coolant flow independent of engine speed [2], [13]

1.4.3 Split-Cooling System with Controlled Bypass

This concept separates the coolant flow control between the cylinder head and the engine block using an electronic valve and bypass line. During a cold start, coolant flow is prioritized through the cylinder head while the engine block flow is temporarily restricted

The engine temperature increases, the valve gradually opens the block circuit, and Radiator path to provide full cooling under high-load Conditions [13], [3],[1], [7].

1.5 Final Concept Selection

After evaluating performance potential, controllability, system complexity, and feasibility, Concept 2, combined with Concept 3, is selected as the final solution.

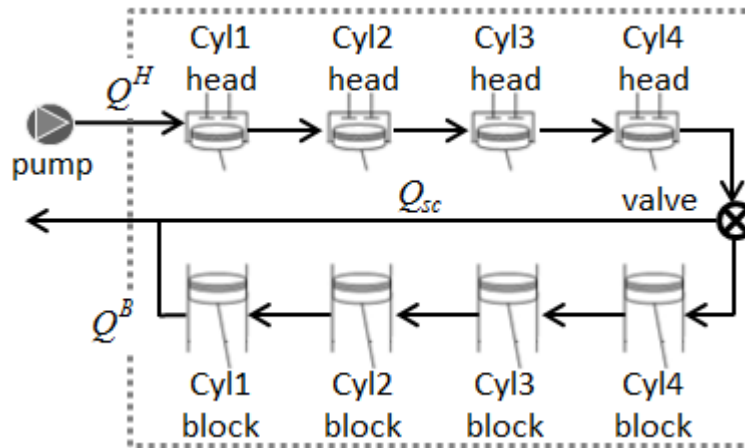


Figure 1.5.1. Final Cooling System Concept

The proposed system integrates an electrically controlled water pump with a split cooling architecture and a controlled bypass mechanism to achieve superior thermal management.

The cooling system is designed based on a head-first cooling strategy, where coolant flow is prioritized through the cylinder head before reaching the engine block. This approach targets the region with the highest thermal load, resulting from combustion, and significantly improves heat removal from critical areas [7], [13].

At engine start-up, the system operates in a bypass mode. The electronic control valve restricts coolant flow to the engine block and radiator, directing most of the coolant toward the cylinder head. This allows the engine to reach optimal operating temperature faster, improving efficiency and reducing cold-start emissions.

As the engine temperature increases, the electronic control valve gradually opens the main cooling path. Coolant is then distributed between the cylinder head, engine block, and radiator according to real-time thermal demand and engine load conditions.

The electric water pump plays a key role by controlling the coolant flow rate independently of engine speed. This enables adaptive cooling performance, ensuring sufficient heat removal under high-load conditions while minimizing unnecessary energy consumption during low-load operation [2], [13]. The coolant, after absorbing heat from the cylinder head and engine block, flows to the radiator, where heat is rejected to the ambient air. The integration of controlled flow distribution and variable pump operation ensures improved thermal stability across the engine.

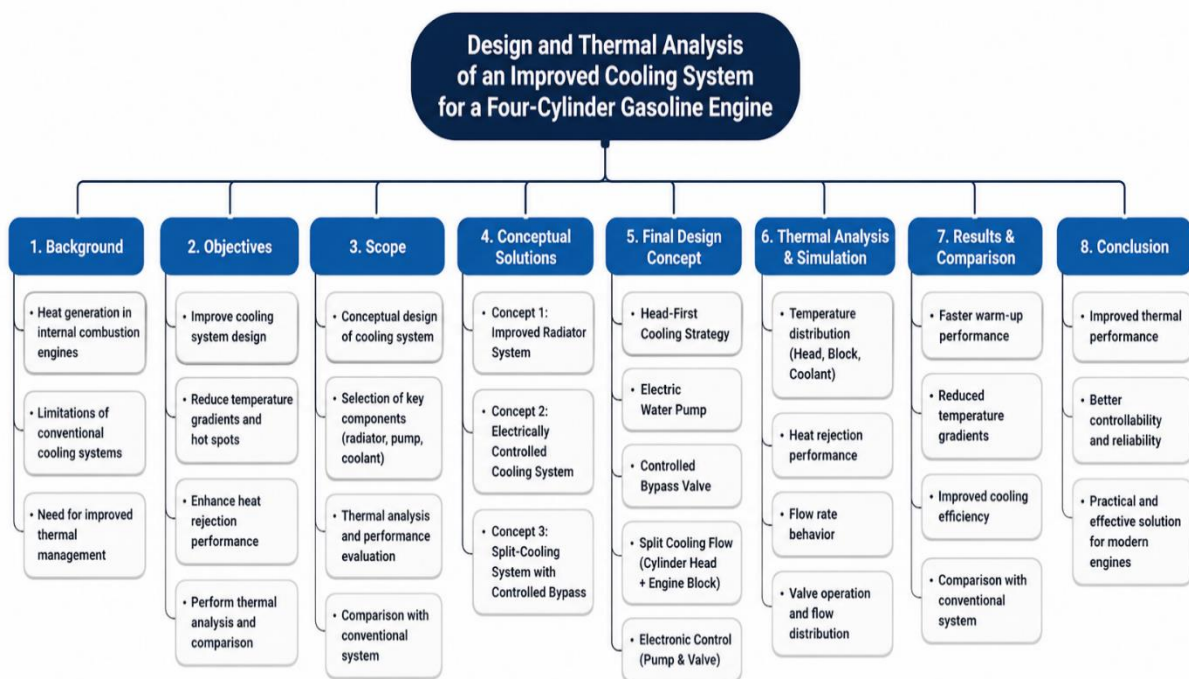
Compared to a conventional single-loop cooling system, the proposed configuration offers

several advantages. It reduces temperature gradients, minimizes hot spots in the cylinder head, accelerates engine warm-up, and enhances overall cooling efficiency.

By combining head-first cooling, split flow paths, and electronic control, the system achieves a balanced and optimized thermal performance, making it a practical and effective solution for modern four-cylinder gasoline engines operating under transient and high-load conditions.

Section 2 – Work Plan

2.1 Work Breakdown Structure (WBS)



2.2 Project Work Breakdown Structure (WBS)

1. Project Planning

Definition of project objectives, scope, methodology, and overall timeline.

2. Literature Review

Comprehensive review of previous studies and technical literature related to internal combustion engine cooling systems.

3. Conceptual Design

Development of conceptual cooling system configurations and evaluation based on thermal performance, feasibility, and system complexity.

4. Thermal Analysis and Simulation

Execution of thermal analysis using analytical heat transfer calculations and numerical simulations to evaluate:

- Temperature distribution
- Coolant flow behavior
- Heat rejection performance

5. Evaluation and Optimization

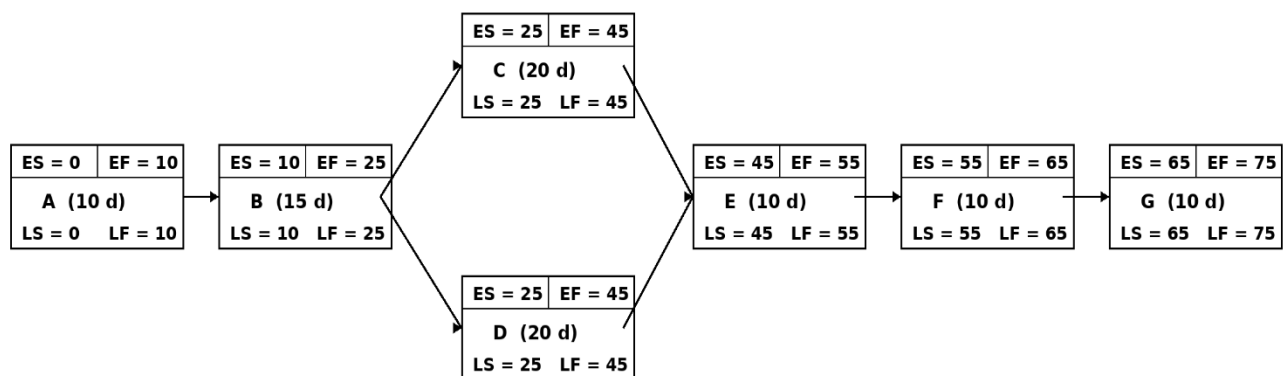
Comparison of the proposed cooling system with a conventional single-loop system. Optimization is performed based on thermal analysis results to improve performance and reliability.

6. Reporting and Presentation

Documentation of the design process, analysis methodology, and results in a final technical report, followed by preparation of a formal presentation.

2.3 Precedence Network (PN):

Activity	Description	Duration (WD)	ES	EF	LS	LF
A	Literature Review	10	0	10	0	10
B	Conceptual Design	15	10	25	10	25
C	Thermal Analysis & Calculations	20	25	45	25	45
D	Simulation	20	25	45	25	45
E	Optimization & Evaluation	10	45	55	45	55
F	Draft Report Preparation	10	55	65	55	65
G	Final Report & Presentation	10	65	75	65	75



Project tasks will follow this sequence:

- Literature Review → Conceptual Design → Thermal Modeling →
Simulation Analysis → Evaluation → Reporting and Presentation

Precedence Network diagram, showing

early start (ES),

early finish (EF),

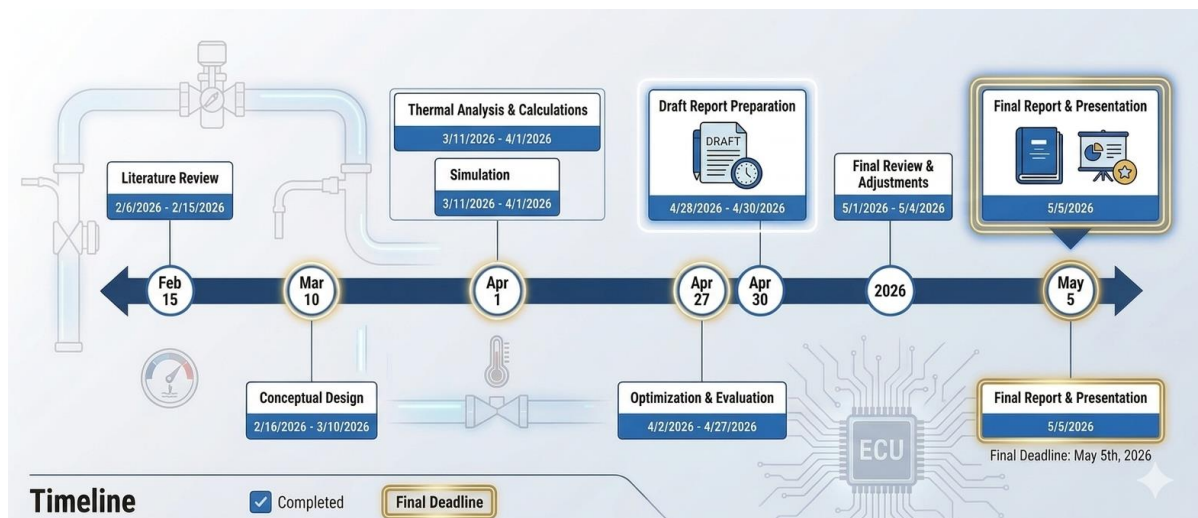
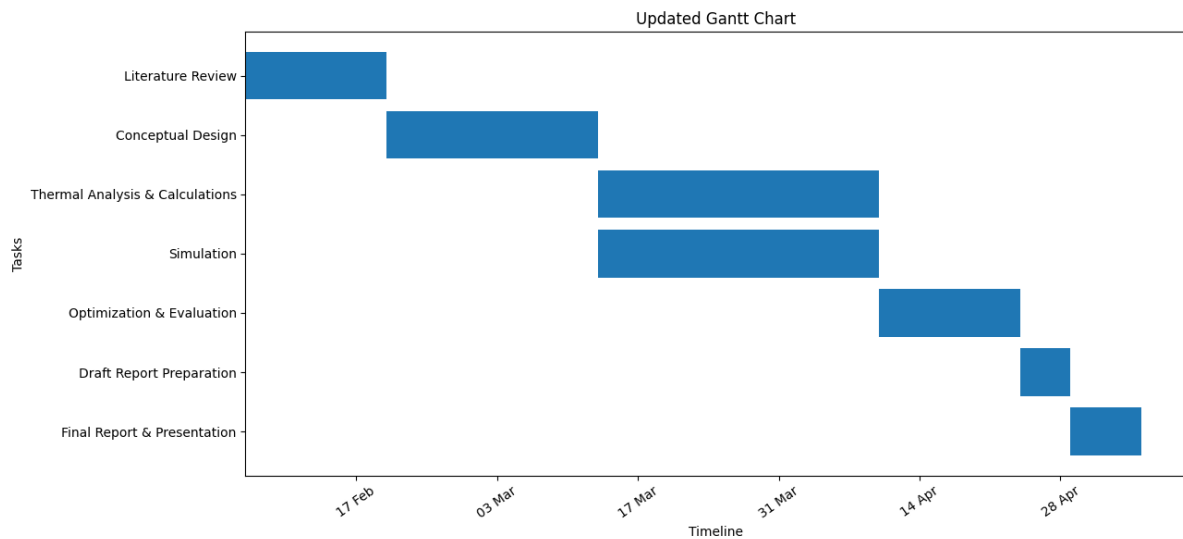
late start (LS),

late finish (LF)

times for each activity.

The total project duration is 75 working days

2.4 Gantt Chart



- The total calendar duration of the graduation project extends from 6 February to 5 May, corresponding to approximately 75 effective working days.
- The project timeline is structured to ensure systematic progress, literature review, conceptual design, thermal analysis, simulation, optimization, and final documentation.

Section 3 – Design Process

3.1 Design Alternatives Evaluation

Each conceptual solution is evaluated based on thermal performance, system complexity, cost, and feasibility.

Concept	Controllability	Reliability	Scalability	Integration
Conventional Cooling System	Low passive control with fixed flow characteristics	Highly proven and robust technology	Limited, difficult to adapt to varying engine loads	Easy compatible with existing engine layouts
Split-Cooling System with Controlled Bypass	Very High independent temperature control for critical components	Medium increased components and control logic	Excellent suitable for high-performance and downsized engines	Complex requires careful system layout and calibration

Table 3.1.1

Conventional Cooling System:

Illustrating coolant circulation through the engine block and radiator using a mechanically driven water pump.

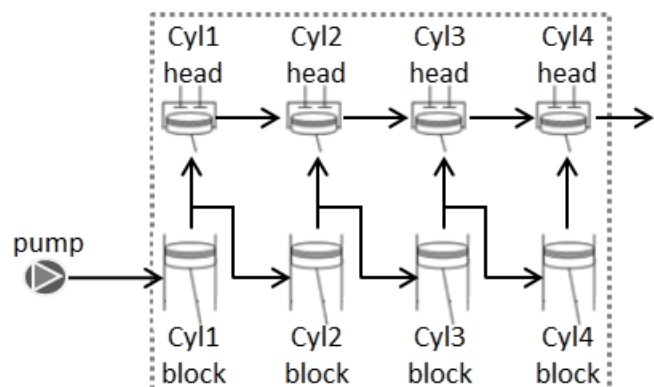


Figure 3.1.1. Conventional Cooling System Layout

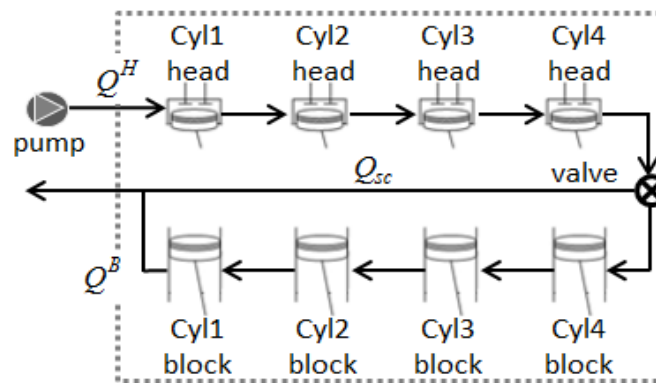


Figure 3.1.2. Split-Cooling System with Controlled Bypass

Split-Cooling System with Controlled Bypass (dual loop): a cooling system with independent cooling circuits for the cylinder head and cylinder block, providing enhanced thermal control and reduced temperature non-uniformity.

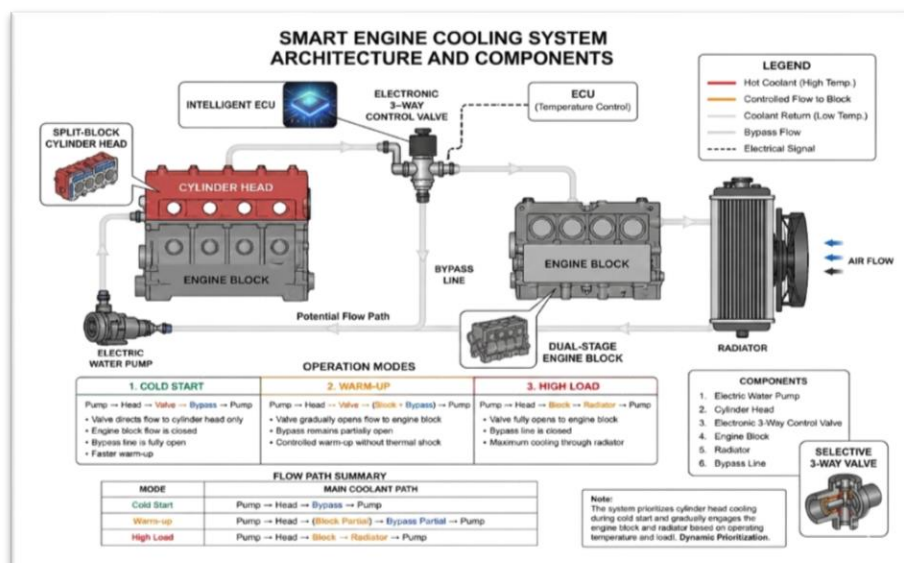


Figure 3.1.3. Cold Start Cooling Operation

This figure shows the layout and operation of the smart engine cooling system. The system uses an electric water pump, a 3-way control valve, and a split-cooling design to control coolant flow between the cylinder head, engine block, and radiator. During operation, the coolant first flows to the cylinder head, which is the hottest part of the engine. The control valve then directs the coolant either through the bypass line or towards the engine block and radiator, depending on engine temperature and load.

At cold start, the coolant flows mainly through the bypass to allow fast warm-up. During normal operation, the flow is distributed between the bypass and the engine block. At high load, the coolant is directed fully to the radiator for maximum cooling. The ECU controls the valve and pump to adjust the system dynamically, ensuring efficient temperature control.

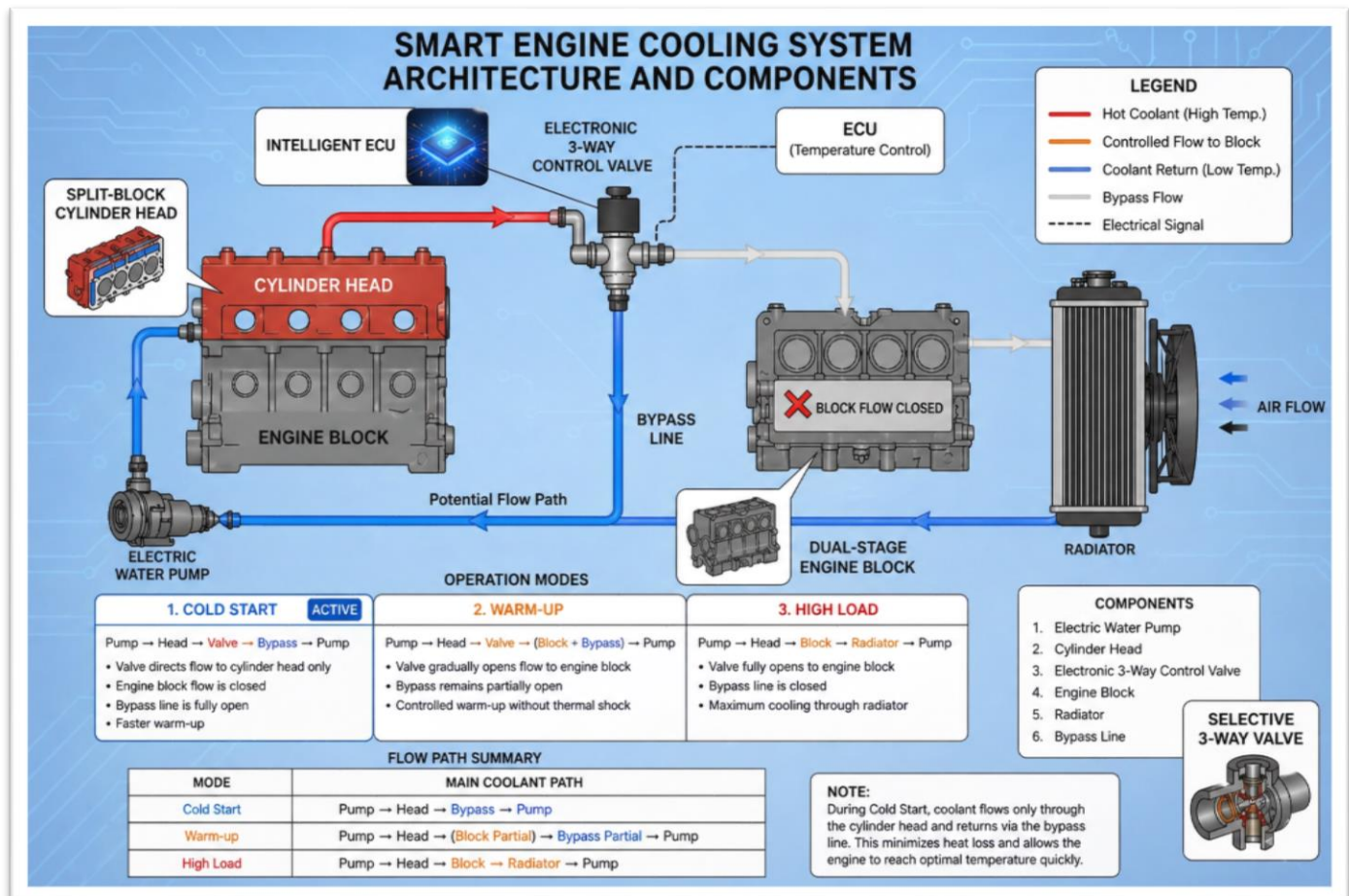


Figure 3.1.4. Cold Start Cooling Operation

This figure shows the operation of the smart cooling system during the cold start condition. At this stage, the coolant is directed mainly to the cylinder head, which is the hottest part of the engine. The flow to the engine block and radiator is closed, and the coolant circulates through the bypass line back to the pump. This configuration reduces heat loss and allows the engine to warm up quickly. The electric pump maintains controlled flow, while the ECU manages the control valve to keep the bypass path fully open. As a result, improved efficiency reduces cold-start emissions.

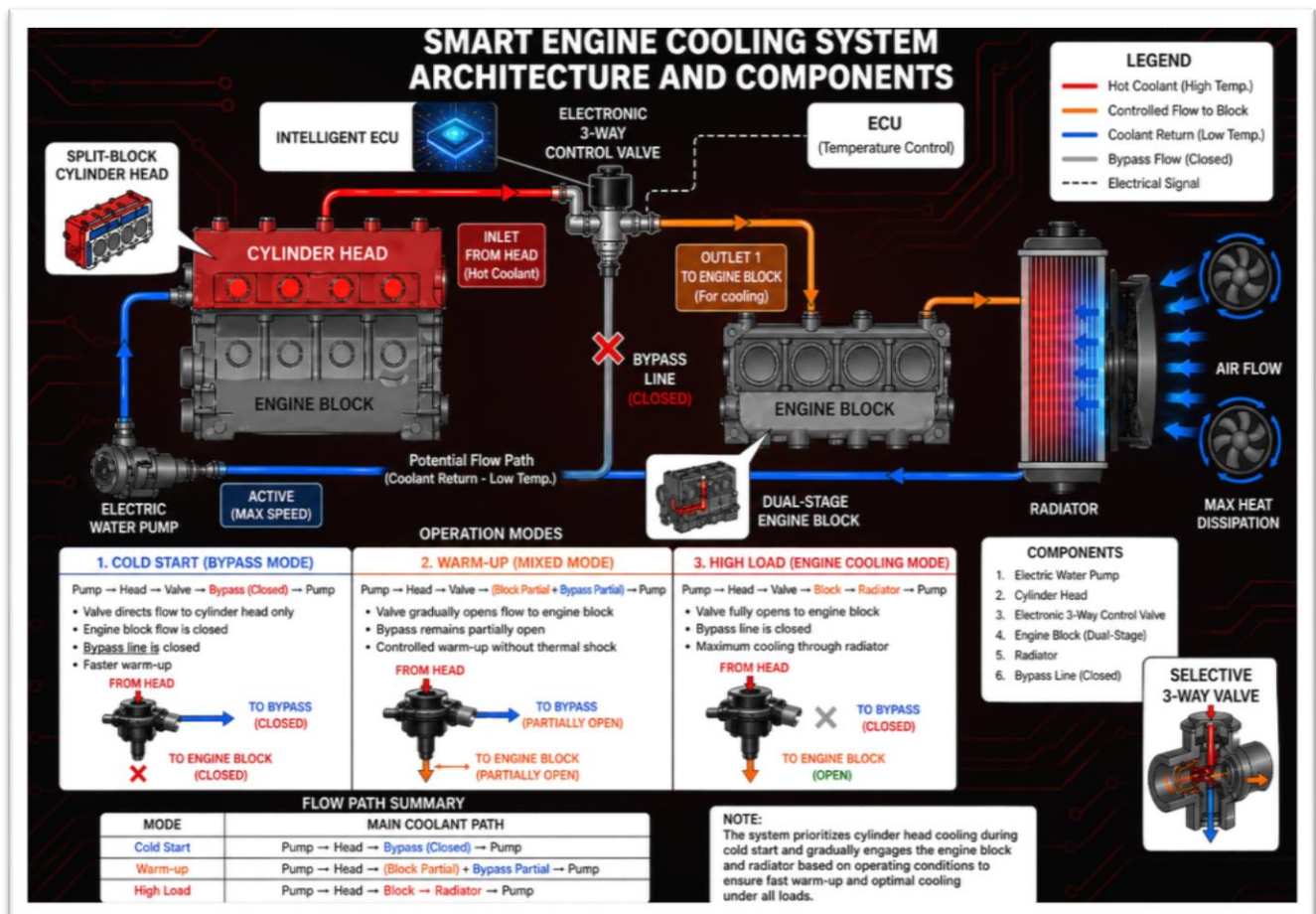


Figure 3.1.5. Cooling System Operation Under Different Conditions

This figure illustrates the operation of the smart engine cooling system under different engine conditions. The system uses an electric water pump and an electronic 3-way control valve to regulate coolant flow between the cylinder head, engine block, radiator, and bypass line.

At high load conditions, coolant flows from the cylinder head to the engine block and then to the radiator, where heat is dissipated. In this case, the bypass line is closed, and the pump operates at maximum speed to ensure efficient cooling.

The control valve, managed by the ECU, directs the coolant flow depending on temperature and engine load, allowing optimal thermal management and improved engine performance.

3.2 Design Decision

The electrically controlled cooling system with enhanced coolant flow management offers the best balance between performance improvement and system feasibility. This design allows adaptive thermal control under different operating conditions and reduces the risk of localized overheating [1], [8], [12].

The selected design prioritizes coolant flow to the cylinder head during cold start conditions to achieve faster warm-up and reduce thermal stress. As the engine temperature increases, the system gradually distributes coolant to the engine block and radiator, ensuring balanced temperature control and efficient heat rejection.

This design achieves an optimal balance between thermal performance, system controllability, and practical feasibility, making it a suitable solution for modern four-cylinder gasoline engines.

3.3 CAD Model and System Design

The final cooling system design is developed using a 3D CAD (Fusion 360) modeling approach [11] to visualize the system layout and ensure proper integration of all components.

The model represents the complete cooling system, including the cylinder head, engine block, radiator, electric water pump, and coolant flow paths. Special attention is given to the routing of coolant passages to ensure efficient heat removal from critical regions.

The CAD design reflects the split-cooling configuration, where the coolant flow is directed first to the cylinder head and then distributed to the engine block through a controlled bypass system.

The use of CAD modeling allows a better understanding of component arrangement, flow paths, and system integration



Figure 3.3.1. Overall Cooling System Layout



Figure 3.3.2. Alternate View of Cooling System Integration

3.3.1 Radiator

The radiator is a key component of the proposed cooling system, responsible for dissipating the heat absorbed by the coolant from the engine. In this design, an aluminum radiator equipped with dual electric cooling fans is used to enhance heat rejection performance under various operating conditions.

The radiator operates by transferring heat from the hot coolant to the surrounding air through convection [4], [14]. The inclusion of dual fans ensures sufficient airflow across the radiator core, particularly at low vehicle speeds or idle conditions where natural airflow is limited.

The use of an aluminum structure improves thermal conductivity while reducing overall system weight. Additionally, the radiator is integrated with the cooling loop in a way that supports the dual-loop configuration.

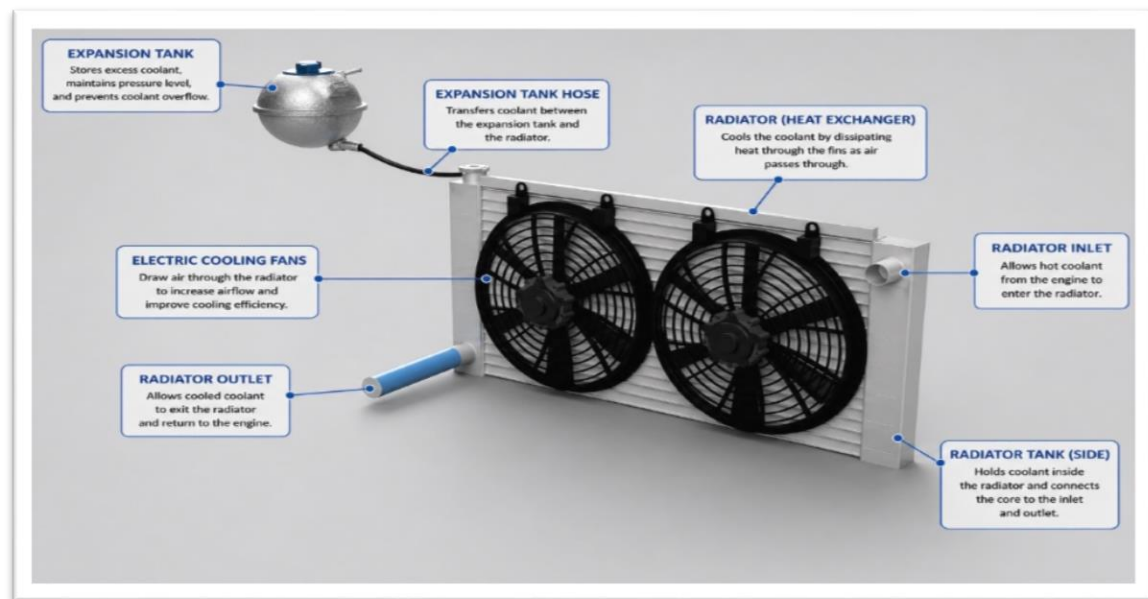


Figure 3.3.3. Radiator with Dual Electric Cooling Fans

3.3.2 Electric Water Pump

The electric water pump plays a critical role in the proposed cooling system by providing active and controllable coolant circulation. Unlike conventional mechanical pumps that are directly driven by engine speed, the electric pump operates independently and is controlled by the engine control unit (ECU) [2], [9].

The ECU continuously monitors key engine parameters such as coolant temperature, engine load, and operating conditions. Based on these inputs, it sends control signals to adjust the pump speed and flow rate in real time. This allows the cooling system to respond dynamically to varying thermal demands.

During cold start conditions, the ECU reduces the pump speed or temporarily limits coolant circulation to accelerate engine warm-up. This helps the engine reach its optimal operating temperature faster, improving fuel efficiency and reducing emissions.

Under normal operating conditions, the pump maintains a moderate and stable flow rate to ensure uniform temperature distribution across the engine components. When the engine is subjected to high load or high-speed operation, the ECU increases the pump speed to enhance coolant circulation and improve heat removal.

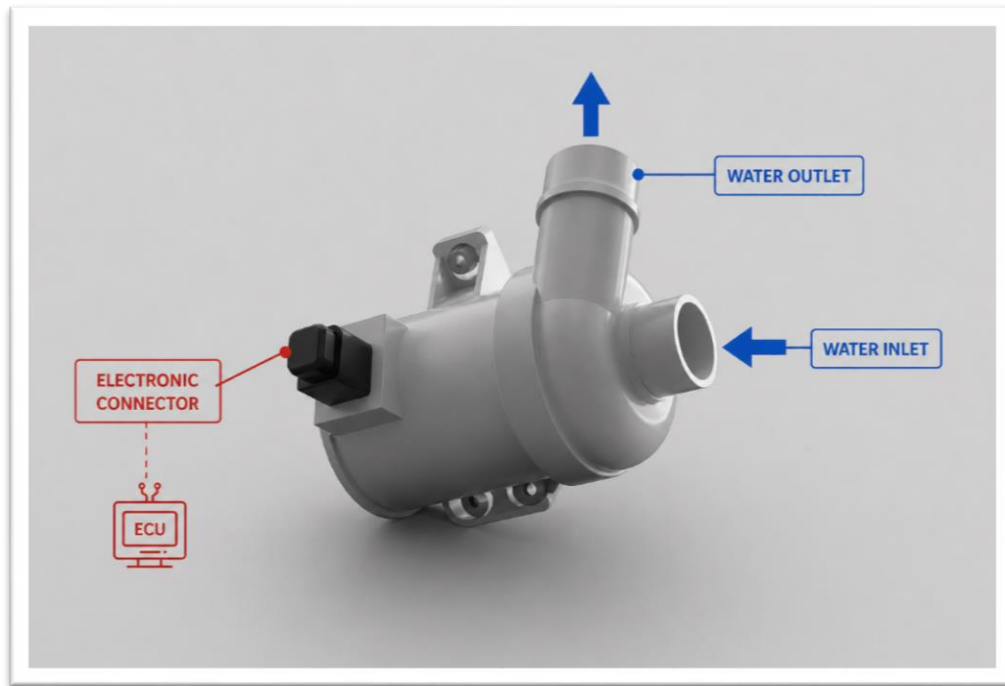


Figure 3.3.4. Electric Water Pump

3.3.3 Electronic Control Valve (Bypass System)

The electronic control valve is a key component in the proposed cooling system for regulating coolant distribution between different flow paths. It enables the implementation of a controlled bypass mechanism, which is essential for achieving the split-cooling strategy [7].

The valve is connected to the engine control unit (ECU), which continuously monitors engine temperature, load, and operating conditions. Based on these inputs, the ECU adjusts the valve position to control the proportion of coolant flowing through the cylinder head, engine block, and radiator.

During cold start conditions, the valve directs most of the coolant flow through the bypass path, limiting flow to the radiator and engine block. This allows rapid heating of the cylinder head and reduces thermal losses, improving engine efficiency.

As the engine temperature increases, the valve gradually opens the main flow path, allowing coolant to pass through the engine block and radiator. This ensures balanced temperature distribution and prevents overheating.

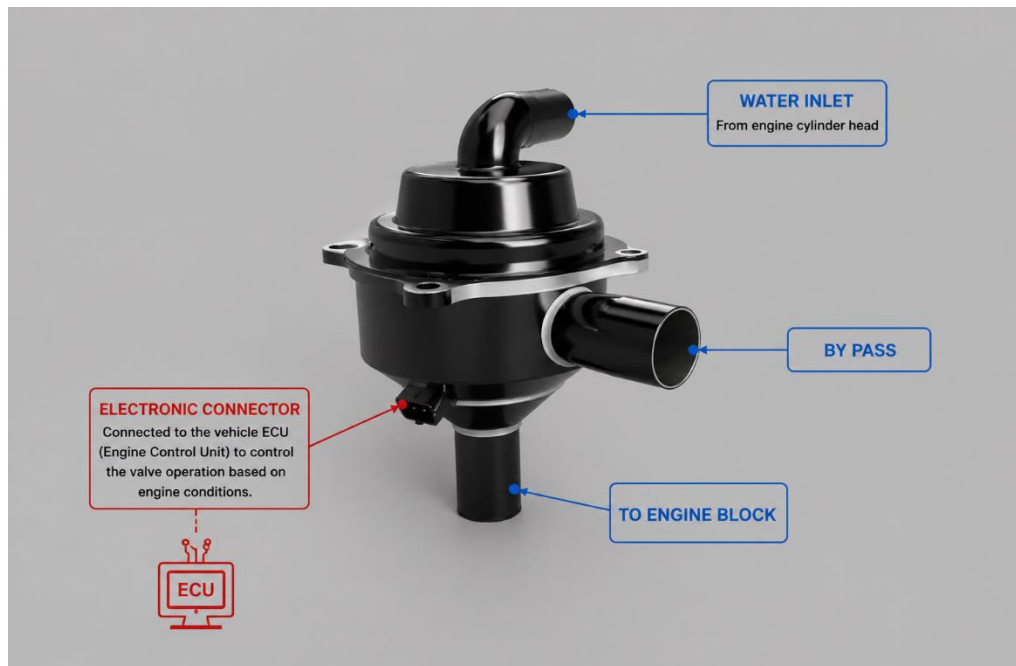


Figure 3.3.5. Electronic Control Valve for Coolant Flow

3.3.4 Pipe Connectors and Fittings

The figure illustrates the coolant flow distribution within the proposed split-cooling system. The system is designed to control the flow of coolant between the cylinder head, engine block, radiator, and bypass path using an electronically controlled valve.

The coolant is first directed toward the cylinder head, where the highest thermal load is generated due to combustion processes. This ensures effective cooling of critical regions and prevents the formation of local hot spots. From the control valve, the coolant can either be directed toward the radiator for heat rejection or through the bypass path, depending on the engine operating conditions.

The bypass line allows the coolant to circulate within the engine without passing through the radiator, which is particularly useful during cold start conditions to accelerate engine warmup. As the temperature increases, the system gradually redirects the flow toward the radiator to enhance heat dissipation.

Additionally, a portion of the coolant is directed toward the engine block to remove residual heat and maintain uniform temperature distribution across the engine structure.

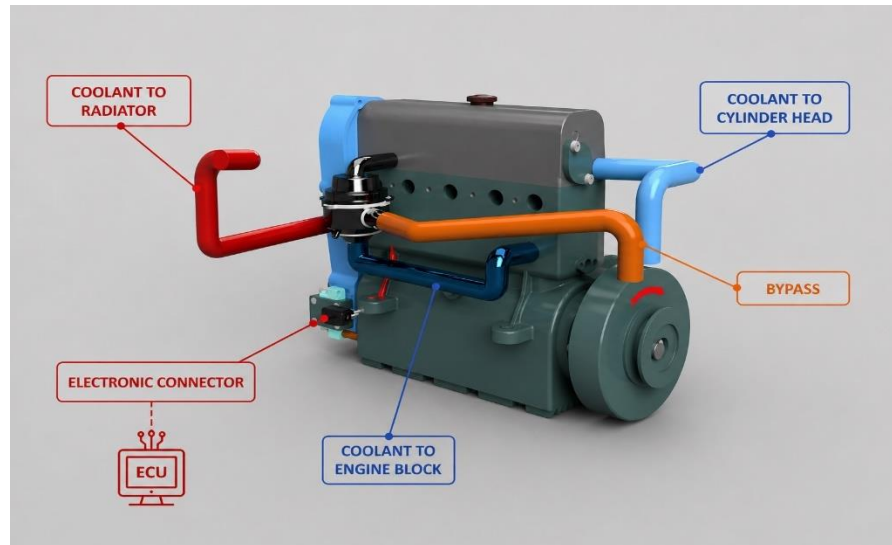


Figure 3.3.6. Cooling Flow Description

The electronic connector linked to the engine control unit (ECU) enables real-time control of the valve position, ensuring dynamic adjustment of coolant flow based on temperature and load conditions. This controlled distribution of coolant flow represents the core functionality of the proposed cooling system, providing improved thermal management, faster warm-up, and enhanced engine reliability compared to conventional systems.

The combined coolant from the radiator and bypass line is directed toward the electric water pump. It enables mixing of hot and cold coolant streams, helping to stabilize temperature and ensure continuous flow within the cooling system.

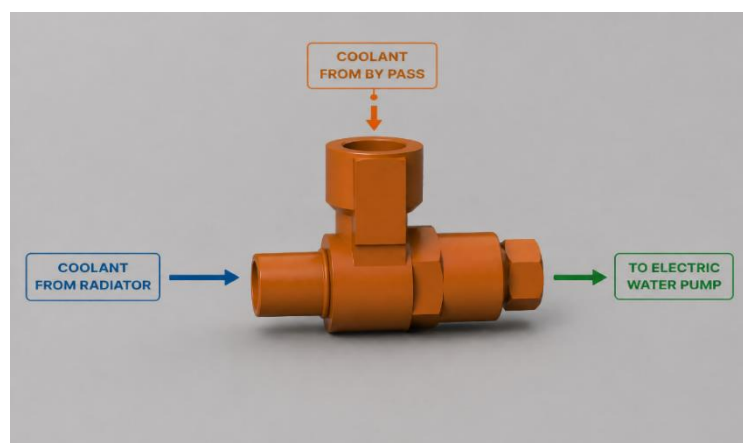


Figure 3.3.7. Coolant Connector

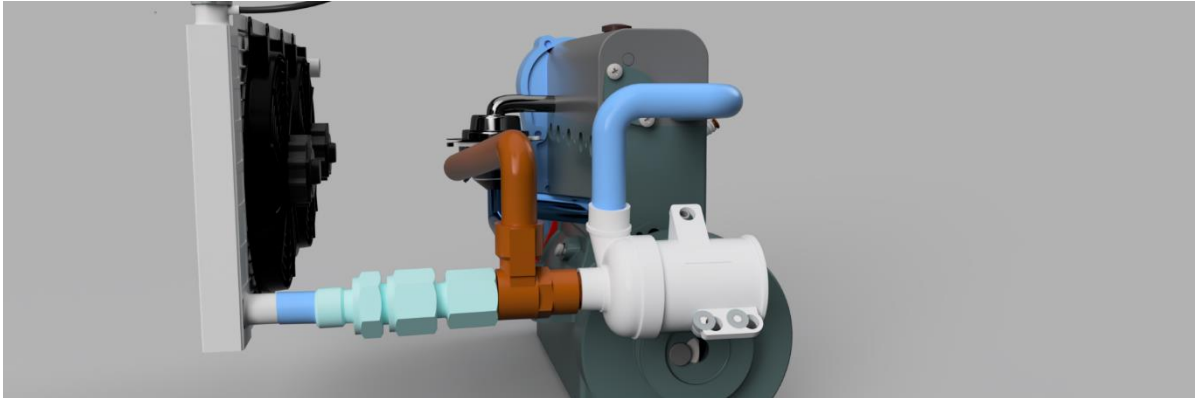


Figure 3.3.8. Integration of Radiator, Connector, and Pump

This figure illustrates the physical integration of the radiator, coolant connector, and electric water pump within the proposed cooling system.

Coolant exiting the radiator, after being cooled, flows through the main outlet pipe toward the connector component. At this junction, the coolant stream may combine with the flow coming from the bypass line, depending on operating conditions.

The connector directs the resulting flow toward the electric water pump, which then circulates the coolant back into the engine. The use of this configuration allows proper mixing of coolant streams before entering the pump, ensuring a more stable inlet temperature and reducing thermal fluctuations within the system.

This arrangement also improves flow continuity and minimizes pressure losses by providing a smooth transition between the radiator and the pump. Additionally, the placement of the electric pump enables active control of the coolant circulation rate, independent of engine speed, enhancing overall cooling performance.

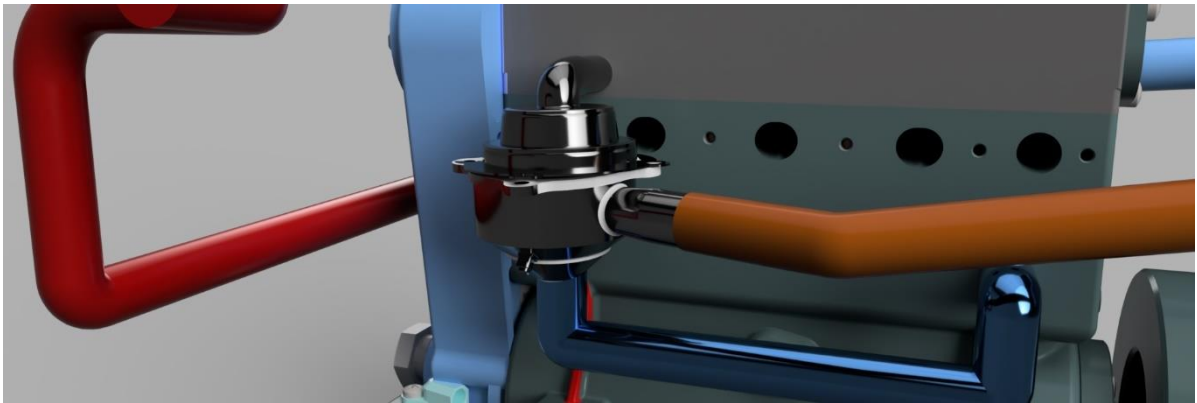


Figure 3.3.9. Coolant Control Valve with Pipes

This figure illustrates the coolant control valve, which plays a key role in actively managing coolant distribution within the cooling system.

Unlike a conventional thermostat, the control valve is electronically actuated and controlled by the Engine Control Unit (ECU). Based on real-time engine parameters such as temperature, load, and operating conditions, the valve adjusts the flow path of the coolant.

During cold-start conditions, the valve prioritizes flow through the bypass circuit, allowing the engine to warm up quickly by limiting heat loss to the radiator. As the engine temperature increases, the valve gradually redirects coolant toward the radiator to enhance heat rejection.

In high-load conditions, the valve can fully open the radiator path to maximize cooling performance, while under light-load conditions, it can partially restrict flow to maintain optimal engine temperature.

This active control strategy improves thermal efficiency, reduces warm-up time, minimizes temperature fluctuations, and enhances overall engine performance and durability.

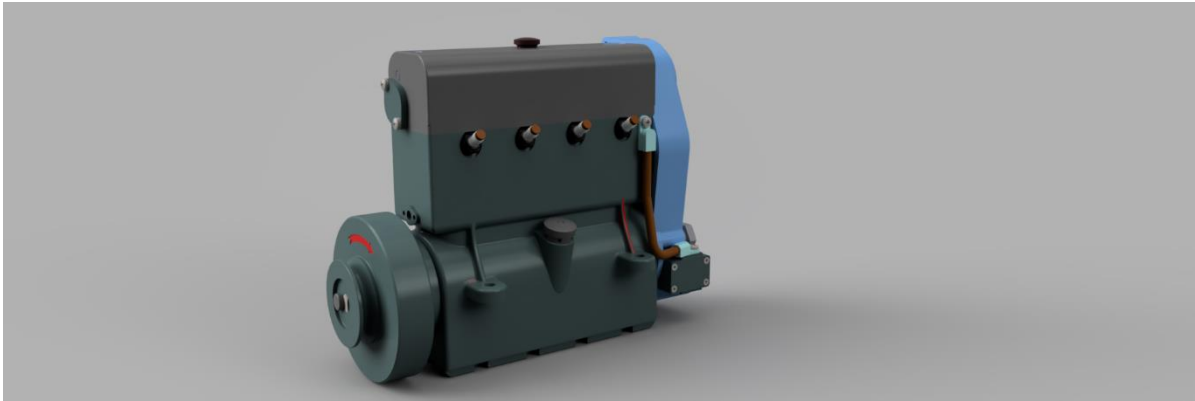


Figure 3.3.10. Assembly of the Four-Cylinder Gasoline Engine

This figure presents a sectional view of the four-cylinder engine, illustrating the internal components and their interaction with the cooling system.

The image highlights the pistons, cylinders, crankshaft, and combustion chamber, which are the primary sources of heat generation during engine operation. As fuel combustion occurs inside the cylinders, a significant amount of thermal energy is produced, particularly in the cylinder head region.

To manage this heat, coolant circulates through internal channels surrounding the cylinder walls and the cylinder head. The cooling system is designed to remove heat efficiently from these critical areas, preventing overheating and maintaining optimal operating conditions.

The cylinder head experiences higher thermal loads due to direct exposure to combustion, while the engine block absorbs and distributes residual heat. This justifies the use of a split-cooling strategy, where cooling of the cylinder head and engine block can be controlled independently.

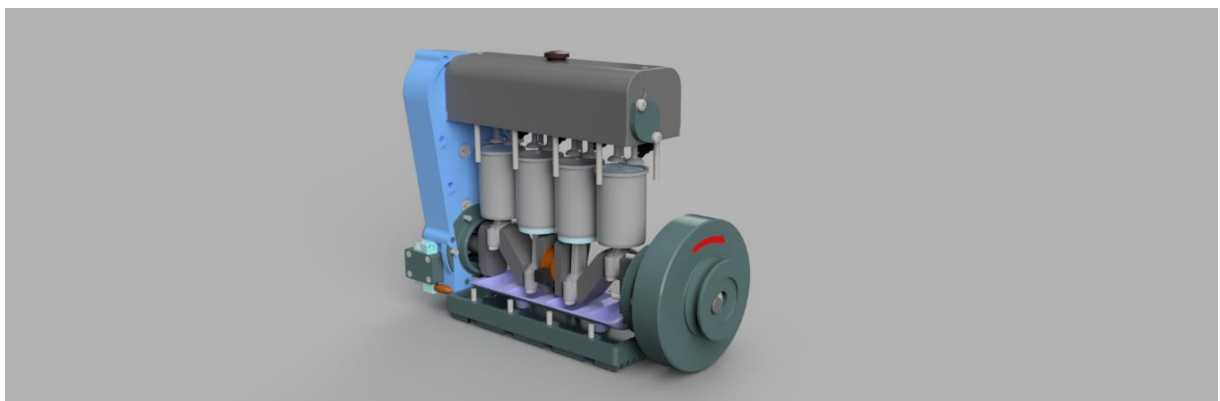


Figure 3.3.11. Internal Structure of the Four-Cylinder Gasoline Engine

Section 4 – Thermal Analysis and Heat Transfer Calculations:

4.1 Thermal Analysis Objective

The thermal analysis is conducted to evaluate the performance of the proposed split-cooling system with an electric water pump and controlled bypass under different engine operating conditions.

The main objective is to determine the system's ability to:

- Maintain optimal engine temperature
- Improve warm-up performance
- Enhance heat rejection under high-load conditions
- Reduce temperature gradients between the cylinder head and engine block

4.2 Operating Conditions

To evaluate system performance, three main operating conditions are considered:

1. Cold Start Condition

- Engine temperature: Low (20–40°C)
- Engine speed: Low
- Load: Minimal

At this stage, the cooling system prioritizes fast warm-up:

- Coolant flows mainly through the bypass circuit
 - Radiator flow is minimized
 - Pump operates at low speed
-

2. Normal Operating Condition

- Engine temperature: Optimal (85–95°C)
- Engine speed: medium
- Load: medium

In this condition:

- Coolant is distributed between engine and radiator
 - Control valve partially opens radiator path
 - Pump adjusts flow rate to maintain stable temperature
-

3. High Load Condition

- Engine temperature: High (95–105°C)
- Engine speed: High
- Load: High

At this stage:

- Radiator flow is maximized
 - Bypass flow is minimized
 - Pump operates at high speed
 - Cooling priority is given to the cylinder head
-

4.3 Heat Transfer Calculations

The heat removed by the coolant is calculated using:

$$\dot{Q} = \dot{m} C_p (T_{out} - T_{in}) \quad [4],[14]$$

Application to Your System:

In the proposed system:

- **Cylinder Head Loop:**
Receives higher heat load $\rightarrow$ higher $\dot{Q}$
- **Engine Block Loop:**
Lower heat $\rightarrow$ secondary cooling

Thus:

$$\dot{Q}_{Total} = \dot{Q}_{Head} + \dot{Q}_{Block} \quad [4]$$

This supports the use of a split-cooling strategy

4.4 Radiator Heat Rejection

The radiator removes heat from the coolant according to:

$$\dot{Q}_{rad} = \dot{m} C_p (T_{hot} - T_{cold}) \quad [4], [14]$$

4.5 Effect of Electric Water Pump

The electric water pump plays a key role in thermal control:

- **Cold start:** Low flow $\rightarrow$ faster warm-up
- **Normal operation:** Controlled flow $\rightarrow$ stable temperature
- **High load:** High flow $\rightarrow$ maximum cooling

Unlike mechanical pumps, flow rate is independent of engine speed, improving system efficiency [9], [13].

The pump is controlled by the ECU based on real-time temperature feedback.

ELECTRIC WATER PUMP (EWP)



TECHNICAL SPECIFICATIONS (EXAMPLE)

Parameter	Typical Value
Operating Voltage	12 V DC
Flow Rate (max)	120 L/min
Head Pressure (max)	1.5 bar
Power Consumption (max)	150 W
Speed Range	500 – 6000 rpm
Coolant Temperature Range	-40 °C to 125 °C
Weight	~ 0.9 kg

1. DESCRIPTION

The electric water pump is an electronically controlled pump that circulates coolant through the engine and the cooling system. Unlike conventional mechanical pumps, it operates independently of engine speed, allowing precise control of coolant flow according to engine operating conditions.

2. FUNCTION

The pump ensures the circulation of coolant to maintain the optimum temperature of engine components. It enables efficient thermal management by adjusting the flow rate as required during different operating modes (cold start, normal operation, high load, etc.).

3. KEY FEATURES

- Electronically controlled speed for precise flow regulation.
- Independent of engine speed for improved control and efficiency.
- High efficiency and low power consumption.
- Compact and lightweight design.
- Enables advanced cooling strategies such as split-cooling.
- Quick response to changing thermal demands.

4. ADVANTAGES

- Improves engine warm-up and overall thermal performance.
- Reduces fuel consumption and emissions.
- Enhances component life through better temperature control.
- Supports advanced thermal management and emissions reduction strategies.

Figure 4.5.1. Effect of Electric Water Pump

4.6 Effect of Control Valve and Bypass

The control valve regulates coolant distribution:

- During warm-up → bypass ON, radiator OFF
- During normal operation → partial distribution
- During high load → radiator ON, bypass OFF

This allows:

- Reduced heat loss during warm-up
- Improved heat rejection when needed
- Smooth temperature control

The valve operates as a mixing and distribution unit, combining hot and cold coolant streams.

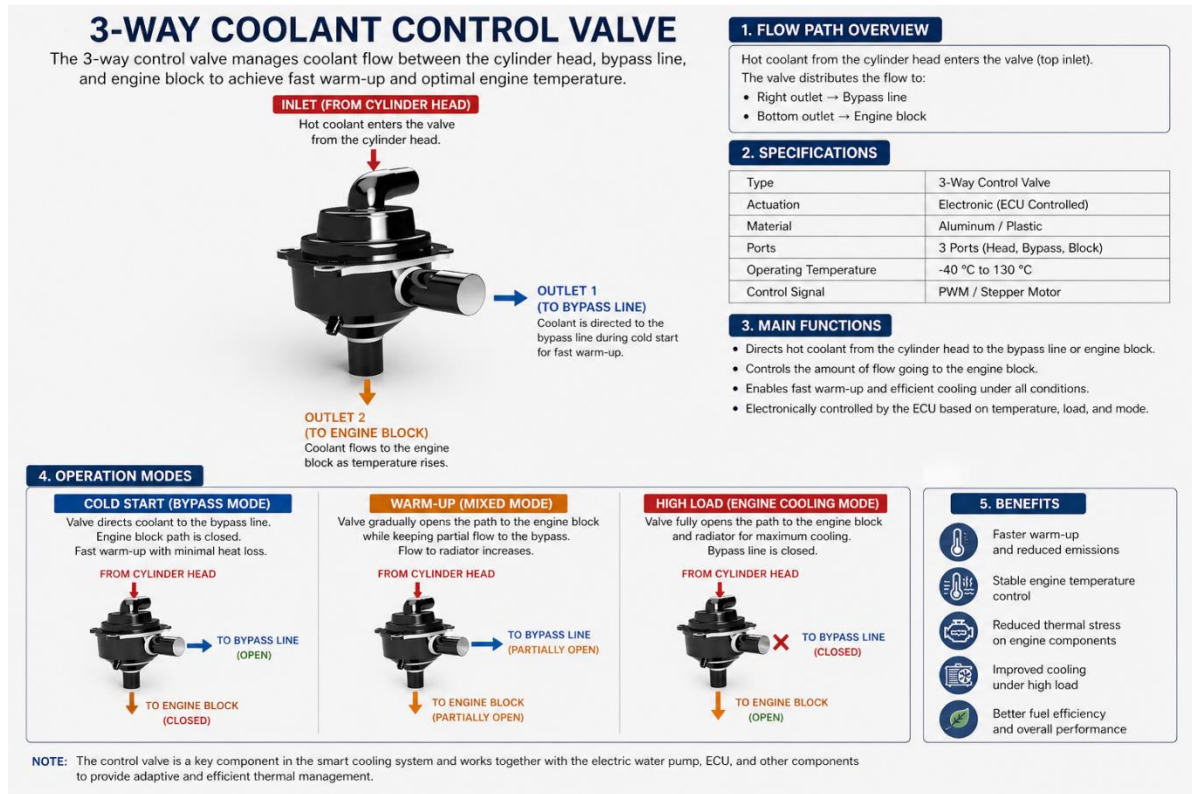


Figure 4.6.1. Control Valve and Bypass Operation

4.7 MATLAB script developed for thermal analysis and coolant flow simulation

As shown in Figure 4.7.1, the engine speed and load profiles used in the MATLAB script-based thermal simulation represent realistic operating conditions, including cold start, normal operation, and high-load conditions.

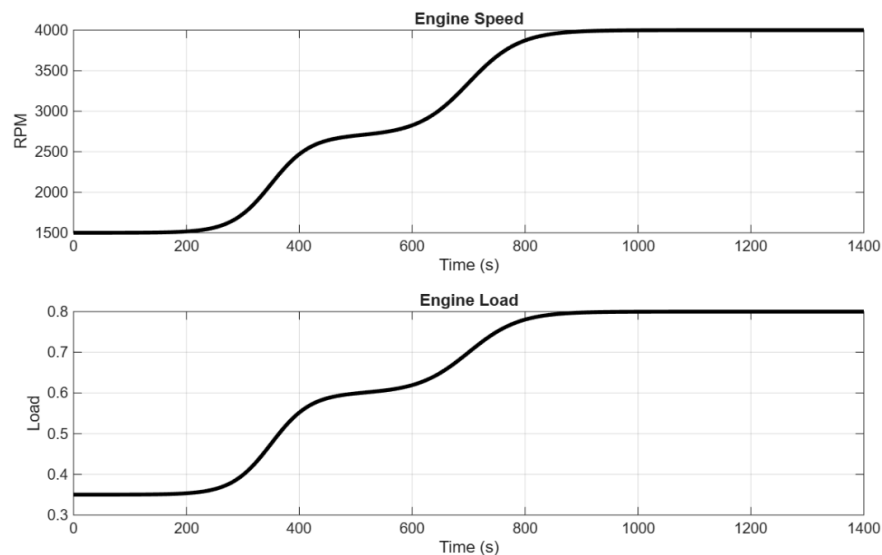


Figure 4.7.1. Engine Speed and Load Profile

The engine operating conditions are gradually increased to simulate real driving. At the beginning, the engine operates under low speed and load conditions, representing a cold start. As time progresses, both engine speed and load increase smoothly until reaching high-load conditions.

The engine speed rises from approximately 1500 RPM to nearly 4000 RPM, while the engine load increases from about 0.35 to 0.80. These changes represent the transition from low thermal demand to high-load engine operation.

The gradual increase in speed and load allows evaluation of the cooling system performance under different operating conditions, including warm-up, normal operation, and high thermal stress conditions.

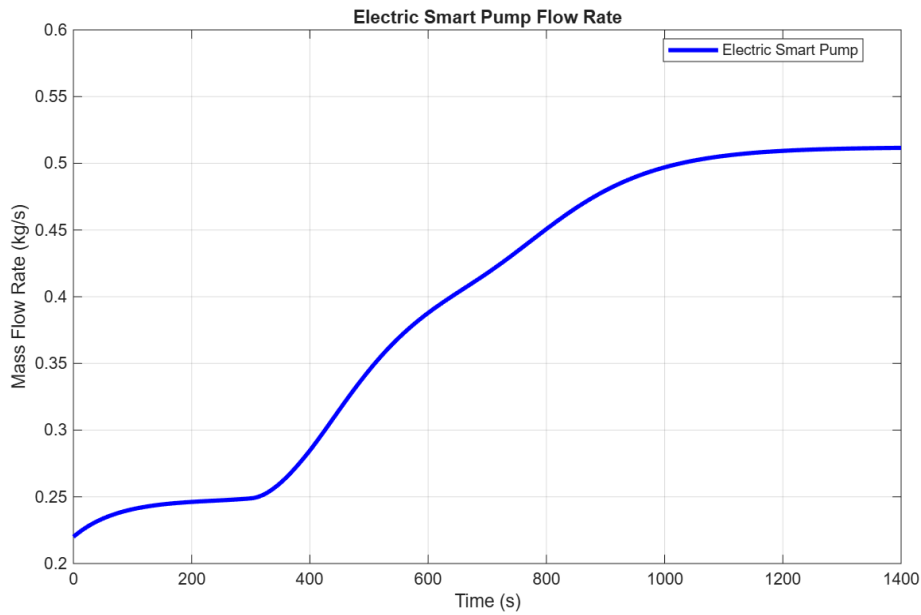


Figure 4.7.2. Electric Water Pump Flow Rate

The electric water pump adjusts its flow rate based on engine operating conditions. At low temperatures, the pump operates at a low flow rate to accelerate engine warm-up.

As engine speed and load increase, the flow rate gradually rises, reaching its maximum value under high-load conditions.

The pump behavior in the MATLAB script is controlled using the following relation:

$$\dot{m}_{pump} = 0.22 + 0.08L + 0.30D_T$$

The coefficients used in the control equation were selected to represent the relative influence of engine load and thermal demand on coolant flow rate within the simplified MATLAB thermal model.

Under high-load operating conditions, the flow rate reaches approximately 0.50–0.52 kg/s, indicating maximum cooling demand. This increased coolant circulation improves heat transfer between the engine and the radiator, helping maintain stable engine temperatures and preventing overheating.

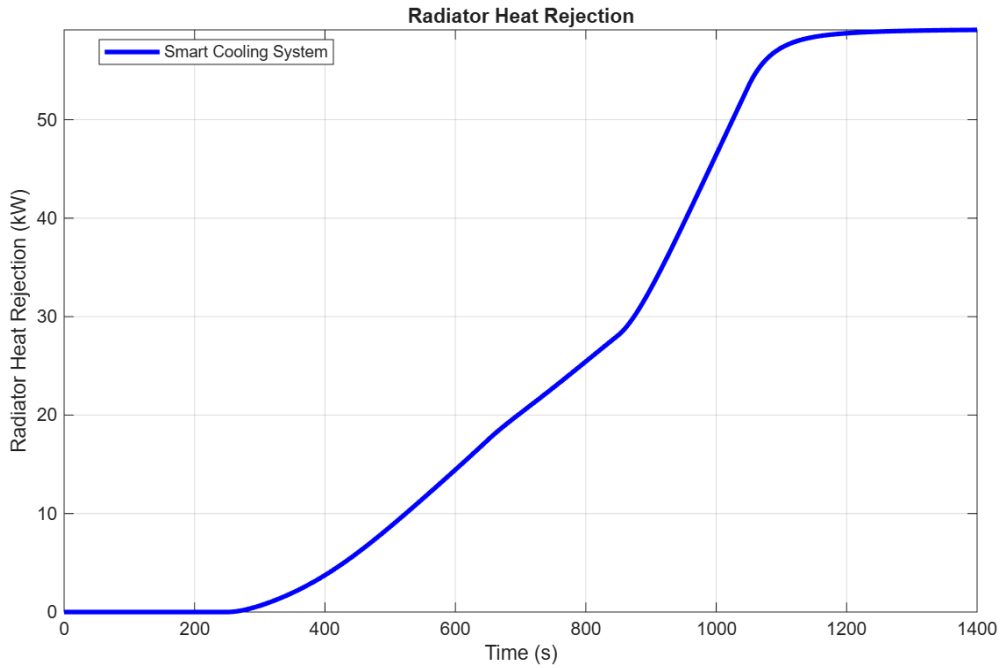


Figure 4.7.3. Radiator Heat Rejection

At the initial stage, heat rejection is minimal since the coolant is mainly circulating through the bypass loop. As the engine temperature increases and the radiator path opens, the heat rejection rises significantly. Under high load conditions, the radiator reaches maximum heat dissipation, confirming its effectiveness in removing excess thermal energy from the system.

This figure illustrates the radiator heat rejection performance of the proposed smart cooling system during the MATLAB thermal simulation. The radiator heat rejection is calculated using the following relation:

$$\dot{Q}_{rad} = \alpha \cdot \frac{\dot{m}}{\dot{m}_{max}} \cdot c_p \cdot (T_{cool} - T_{amb})$$

At high engine loads, the radiator reaches its maximum heat dissipation capacity (approximately 55–57 kW), ensuring effective removal of excess thermal energy and preventing overheating.

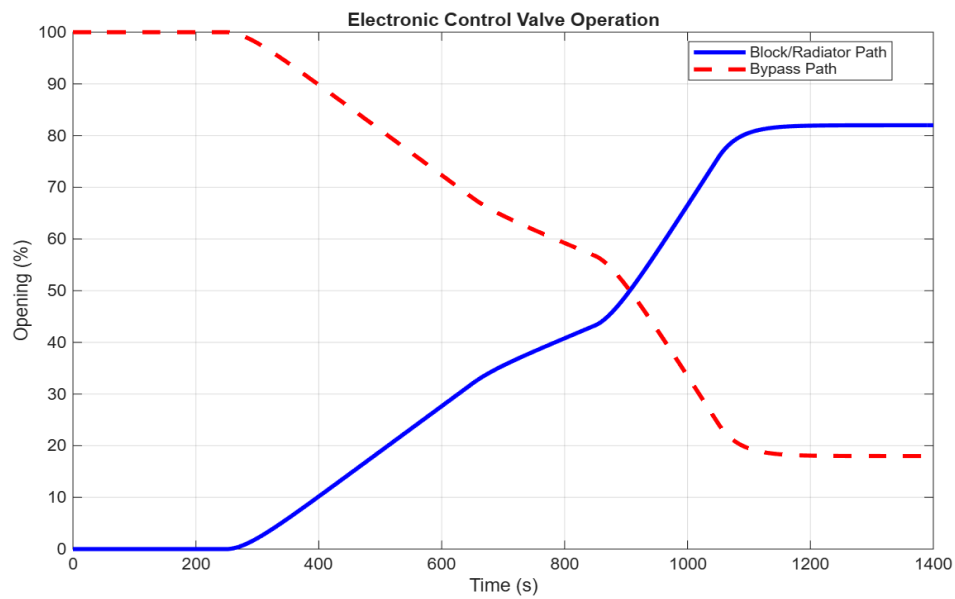


Figure 4.7.4. Control Valve Operation

Control valve opening percentage for radiator and bypass paths.

At a cold start, the bypass path is fully open while the radiator path is closed, allowing rapid engine warm-up. As the engine temperature increases, the valve gradually redirects coolant toward the radiator. Under high load conditions, the radiator path becomes dominant while the bypass is minimized. This dynamic control ensures efficient thermal management across all operating conditions.

At high-load conditions, the radiator path becomes dominant (80% open), while the bypass flow is significantly reduced (20%). This ensures maximum heat removal, especially from critical components such as the cylinder head, where thermal loads are highest.

The results demonstrate that the control valve plays a crucial role in adaptive thermal management by continuously adjusting coolant flow paths according to engine demand. This intelligent flow control improves system efficiency, reduces temperature fluctuations, and enhances overall engine reliability.

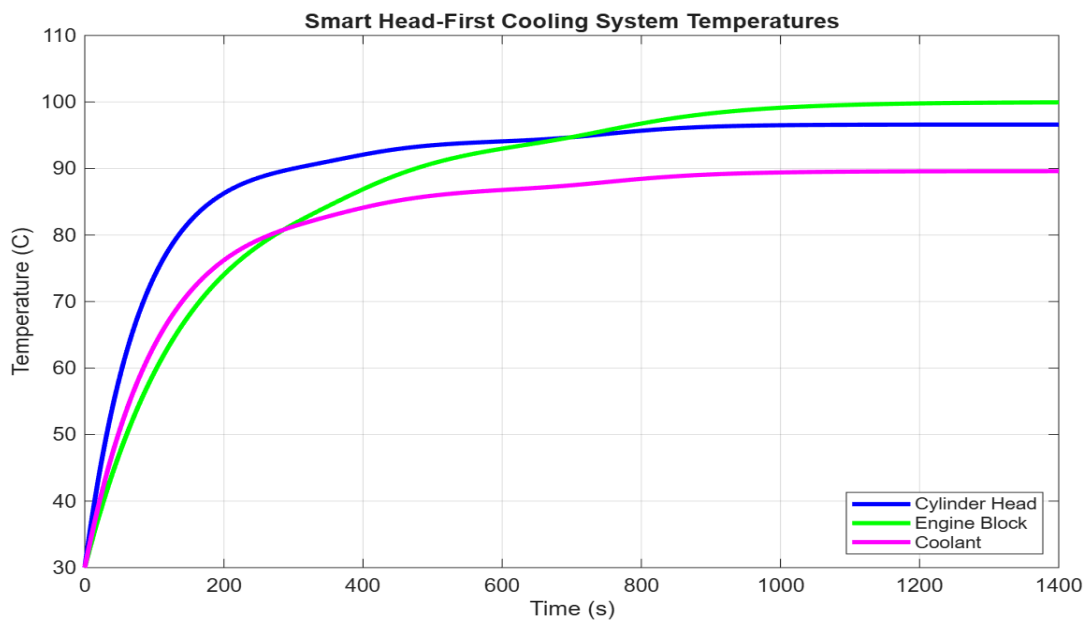


Figure 4.7.5. Temperature Distribution

Temperature variation of the cylinder head, engine block, and coolant.

The figure illustrates the temperature evolution of the main engine components and coolant during the simulation. At the beginning (cold start condition), all temperatures increase rapidly as the engine warms up. The cylinder head temperature rises faster than the engine block due to its direct exposure to combustion heat.

$$T(t) = T_{target} - (T_{target} - T_0)e^{\frac{-t}{\tau}}$$

The equation describes how the temperature gradually approaches the target temperature instead of changing instantly. This behavior represents the real thermal inertia of engine components and coolant flow.

As the system approaches normal operating conditions, the rate of temperature increase slows down, and the temperatures begin to stabilize. The engine block temperature reaches the highest steady-state value (**approximately 99–100°C**), while the cylinder head stabilizes at approximately (**96–97°C**). The coolant temperature remains lower (**88–90°C**), indicating effective heat absorption and transport away from the engine.

The difference between the cylinder head and engine block temperatures is relatively small, demonstrating the effectiveness of the split-cooling strategy in reducing temperature gradients across engine components. This is critical for minimizing thermal stresses and

improving engine durability.

Under high-load conditions, all temperatures increase slightly but remain within safe operating limits. The cooling system successfully maintains thermal stability by increasing coolant flow and heat rejection through the radiator.

Overall, the results confirm that the proposed cooling system provides efficient and controlled temperature regulation, preventing overheating while ensuring optimal engine performance.

4.8 Comparison with Conventional Cooling System

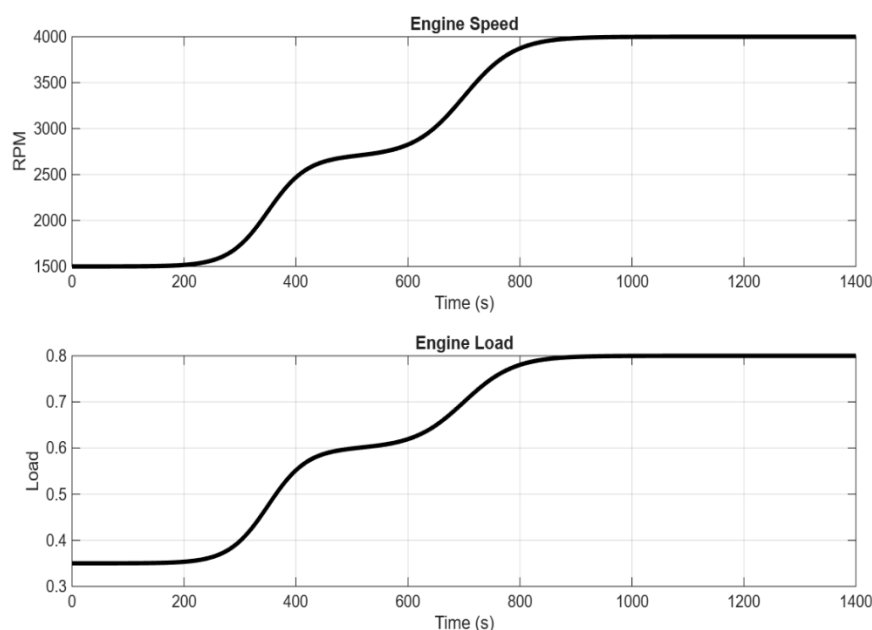


Figure 4.8.1. Engine Speed and Load Profile

The thermal behavior of the proposed cooling system was simulated using MATLAB [10]. A dynamic model was developed to capture the variation of engine speed, load, coolant flow rate, and heat transfer over time. This approach enables accurate evaluation of system performance under realistic operating conditions.

Engine Speed Profile (RPM)

The engine speed is varied to simulate real driving conditions:

- Initial (cold start): **1300 RPM**

- Mid operation: **2800 RPM (at 500 s)**
- High load: **4200 RPM (at 900 s and above)**

This gradual increase represents acceleration from idle to high-speed operation.

Engine Load Profile

The engine load follows a realistic progression:

- Cold start: **0.30**
- Normal operation: **0.55 – 0.65**
- High load: **0.80**

This allows evaluation of the cooling system under increasing thermal demand.

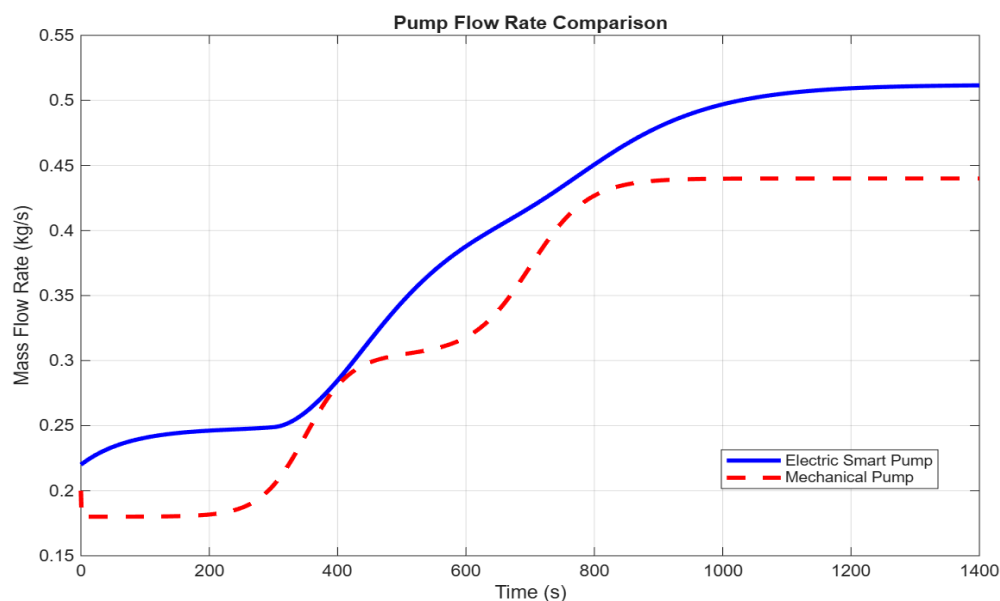


Figure 4.8.2. Pump Flow Rate Comparison

The pump flow rate plays a critical role in controlling the thermal behavior of the engine cooling system. In this study, a comparison is made between a conventional mechanical pump and a smart electric water pump.

The mechanical pump is directly driven by the engine, meaning that its flow rate is proportional to engine speed. As a result, at low engine speeds (cold start), the coolant flow rate is relatively low, which may lead to uneven temperature distribution and slower system response. As engine speed increases, the flow rate rises, but without precise control

over thermal requirements.

In contrast, the smart electric water pump operates independently of engine speed [2], [13] and is controlled electronically [2], [9]. This allows the pump to adjust the coolant flow rate based on real-time engine conditions such as temperature and load.

During cold start, the electric pump provides a moderate flow rate to support faster warm-up without excessive heat loss. Under normal operation, it maintains a controlled flow to stabilize engine temperature. At high load conditions, the pump significantly increases the flow rate to enhance heat removal, especially from critical components such as the cylinder head.

Overall, the smart pump delivers a more flexible and efficient cooling strategy, ensuring optimal thermal management across all operating conditions [1], [12].

Smart Electric Pump:

- Initial: **0.22–0.25 kg/s**
- Normal: **0.35–0.45 kg/s**
- High load: **0.50–0.52 kg/s**

Mechanical Pump:

- Initial: **0.10–0.15 kg/s**
- Normal: **0.30–0.40 kg/s**
- High load: **0.42–0.45 kg/s**

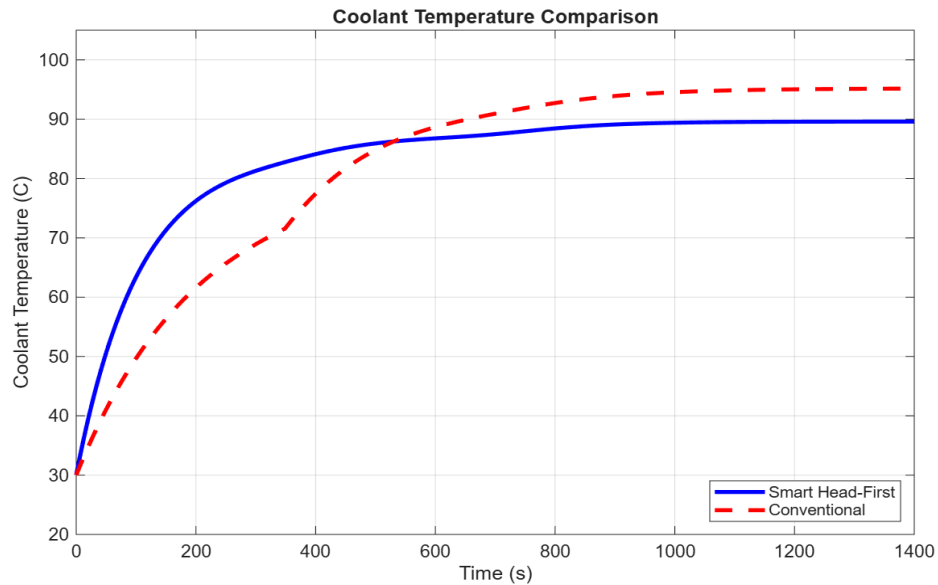


Figure 4.8.3. Coolant Temperature Comparison

The coolant temperature increases gradually during the warm-up phase as it absorbs heat from the engine. In the smart cooling system, the coolant reaches its optimal temperature faster and remains more stable compared to the conventional system.

This stable behavior indicates efficient heat absorption and transfer, helping to maintain proper engine temperature and prevent overheating.

Coolant Temperature

Smart System:

- Final temperature: **88–92°C**

Conventional System:

- Final temperature: **92–96°C**

More stable thermal behavior in the smart system.

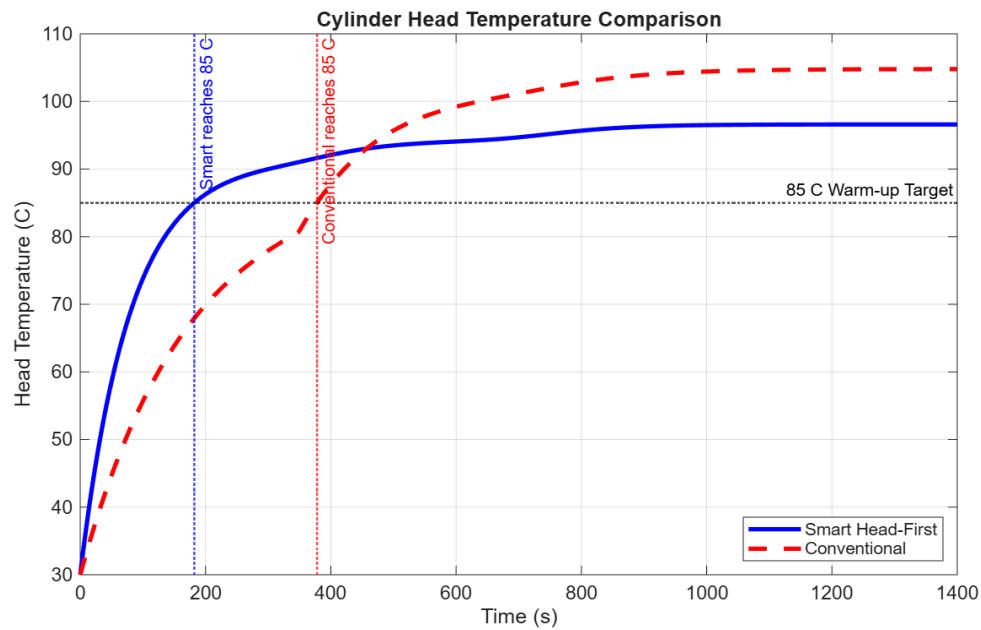


Figure 4.8.4. Cylinder Head Temperature Comparison

The cylinder head temperature increases rapidly during the warm-up phase due to combustion heat. In the smart cooling system, the temperature reaches the optimal range faster compared to the conventional system.

Additionally, the smart system maintains a lower and more stable final temperature, which helps reduce thermal stress and improve engine reliability.

Cylinder Head Temperature

Smart Cooling System:

- Reaches **85°C at ~182 s**
- Final temperature: **~95–98°C**

Conventional System:

- Reaches **85°C at ~378 s**
- Final temperature: **~102–108°C**

Faster warm-up and lower peak temperature in the smart system.

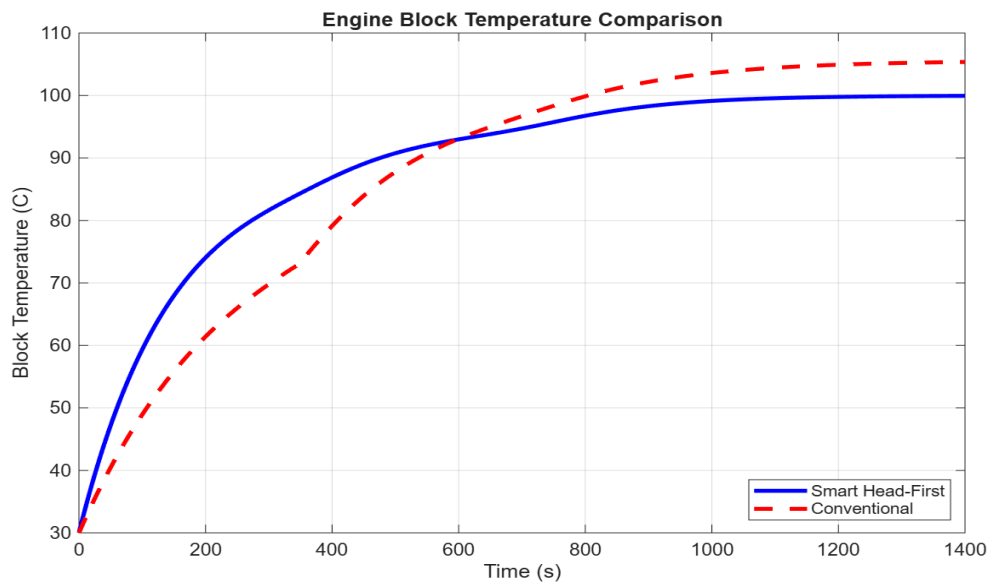


Figure 4.8.5. Engine Block Temperature Comparison

The engine block temperature increases gradually as the engine warms up. In the smart cooling system, the temperature rises faster during the initial phase but stabilizes at a lower value compared to the conventional system.

This indicates better heat control and reduced thermal stress on the engine block, which improves overall engine durability and performance.

Engine Block Temperature

Smart System:

- Final temperature: **~98–100°C**

Conventional System:

- Final temperature: **~103–105°C**

Reduced thermal stress in the proposed design.

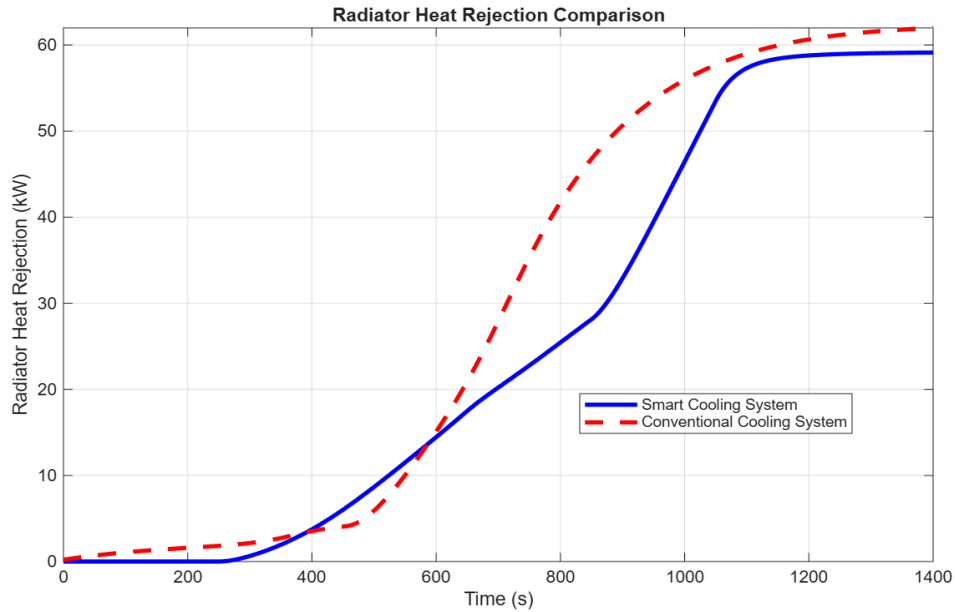


Figure 4.8.6. Radiator Heat Rejection Comparison

The radiator heat rejection increases as the engine temperature rises. In the smart cooling system, heat rejection is low during the warm-up phase to allow faster engine heating. As the engine reaches higher temperatures, the heat rejection increases significantly to remove excess heat.

This controlled behavior improves efficiency by reducing unnecessary cooling at the beginning and providing strong cooling when needed.

Radiator Heat Rejection

Smart System:

- 0–250 s: **0 kW (minimal cooling)**
- 500 s: **10 kW**
- Final: **55–58 Kw**

Conventional System:

- 0–300 s: **1–5 kW**
- 700 s: **30–40 kW**
- Final: **60–62 kW**

Smart system delays cooling to improve warm-up efficiency.

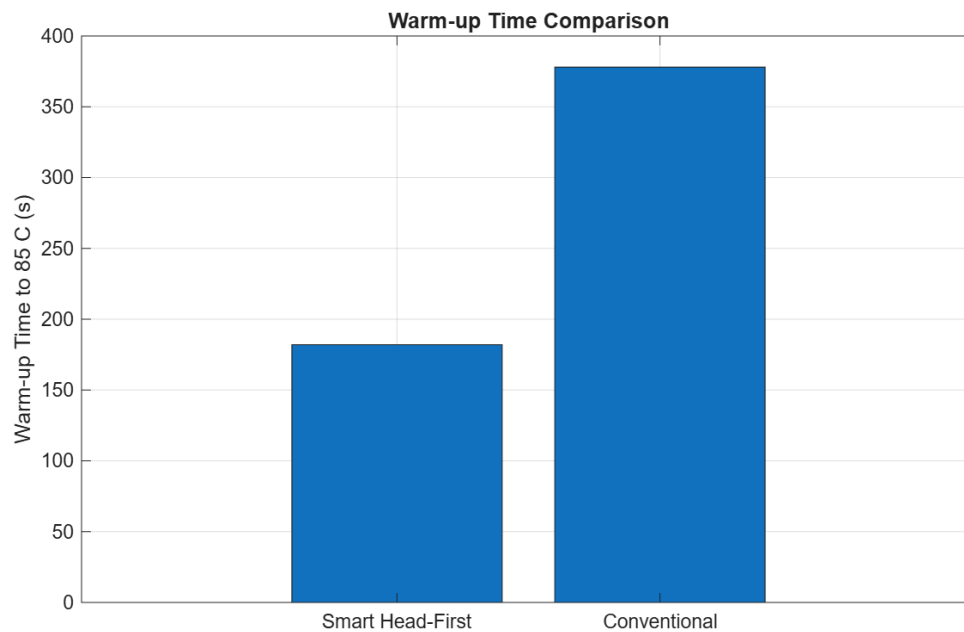


Figure 4.8.7. Warm-up Time Comparison

The warm-up phase represents the time required for the engine to reach its optimal operating temperature. In the smart cooling system, the warm-up process is faster due to the use of the bypass path and controlled coolant flow.

During this phase, the radiator is partially or fully bypassed, reducing heat loss and allowing the engine to heat up more quickly. As a result, the smart system reaches the desired temperature in a shorter time compared to the conventional system.

Warm-up Time

- Smart system: **150–200 seconds**
- Conventional system: **350–450 seconds**

Approx. **50% faster warm-up**

Section 5 – Conclusion

5.1 Overall Conclusion

This project presented the design and thermal analysis of a smart cooling system for a four-cylinder gasoline engine. The proposed system integrates an electric water pump, a controlled bypass valve, and a split-cooling configuration to enhance thermal management under varying engine operating conditions [1], [13].

The results obtained from the MATLAB script-based thermal simulation demonstrate that the smart cooling system significantly improves thermal performance compared to a conventional cooling system. demonstrate that the smart cooling system significantly improves overall performance compared to a conventional cooling system. The system achieves faster warm-up by reducing heat loss during cold start through the use of the bypass path. This allows the engine to reach its optimal operating temperature more quickly, improving efficiency and reducing emissions.

Furthermore, the temperature distribution within the engine is better controlled. The cylinder head and engine block temperatures remain within safe and stable ranges, with reduced temperature gradients between components. This contributes to minimizing thermal stress and improving engine durability.

The use of an electric water pump enables adaptive control of coolant flow, independent of engine speed. This allows the system to respond effectively to different thermal demands, increasing flow rate under high-load conditions to enhance heat removal.

In addition, the control valve plays a crucial role in regulating coolant distribution between the radiator and bypass paths. This ensures smooth transitions between operating modes and provides efficient heat rejection when required.

Overall, the proposed cooling system demonstrates superior thermal performance, improved efficiency, and enhanced reliability. The combination of intelligent flow control and split-cooling strategy makes the system a practical solution for modern internal combustion engines.

5.2 Impact on Engine Performance:

During cold start conditions, the system reduces coolant flow to the engine block and prioritizes the cylinder head, allowing the engine to reach its optimal operating temperature more quickly. This results in reduced friction losses, improved combustion efficiency, and lower fuel consumption.

Under high-load conditions, the system increases coolant flow to prevent overheating, particularly in high-temperature regions such as the cylinder head. This ensures stable engine operation even under demanding conditions.

5.3 Thermal Efficiency and Energy Savings

The use of an electric water pump significantly reduces parasitic losses associated with conventional mechanical pumps. Since the pump operates only when needed and at variable speeds, unnecessary energy consumption is minimized [16].

Additionally, the electronic control valve optimizes coolant routing, preventing excessive circulation and reducing heat losses. This leads to improved fuel efficiency, reduced emissions, and better utilization of engine power.

5.4 Final statement

In conclusion, the proposed cooling system offers an advanced and efficient approach to engine thermal management. By combining electronic control with flexible coolant distribution, the system improves performance, efficiency, and reliability, making it a strong candidate for future automotive applications.

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→ Direct reference for project concept.

APPENDIX A

MATLAB script for the system:

```
clc; clear; close all;
```

```
%% SMART HEAD-FIRST COOLING SYSTEM ONLY
```

```
%% Time
```

```
dt = 1;
```

```
t_end = 1400;
```

```
t = 0:dt:t_end;
```

```
n = length(t);
```

```
%% Initial temperatures
```

```
T_head = zeros(1,n);
```

```
T_block = zeros(1,n);
```

```
T_cool = zeros(1,n);
```

```
T_head(1) = 30;
```

```
T_block(1) = 30;
```

```
T_cool(1) = 30;
```

```
%% Variables
```

```
rpm = zeros(1,n);
```

```
load = zeros(1,n);
```

```
pump = zeros(1,n);
```

```
valve = zeros(1,n);
```

```
bypass = zeros(1,n);
```

```
Qrad = zeros(1,n);
```

```
pump(1) = 0.22;
```

```
valve(1) = 0.00;
```

```
bypass(1) = 1.00;
```

```
%% Response constants
```

```
tau_head = 80;
```

```
tau_block = 145;
```

```
tau_cool = 105;
```

```
tau_pump = 75;
```

```
%% Simulation
```

```
for i = 2:n
```

```
    %% Engine operation: light -> medium -> high
```

```
    load(i) = 0.35 ...
```

```

+ 0.25/(1 + exp(-(t(i)-350)/35)) ...
+ 0.20/(1 + exp(-(t(i)-700)/45));

rpm(i) = 1500 ...
+ 1200/(1 + exp(-(t(i)-350)/35)) ...
+ 1300/(1 + exp(-(t(i)-700)/45));

%% Electronic 3-way control valve
if t(i) < 250
    desired_valve = 0.00;
elseif t(i) < 650
    desired_valve = 0.35*(t(i)-250)/(650-250);
elseif t(i) < 850
    desired_valve = 0.35 + 0.10*(t(i)-650)/(850-650);
elseif t(i) < 1050
    desired_valve = 0.45 + 0.37*(t(i)-850)/(1050-850);
else
    desired_valve = 0.82;
end

valve(i) = valve(i-1) + dt*(desired_valve - valve(i-1))/35;
valve(i) = max(0.00, min(valve(i), 0.85));

bypass(i) = 1 - valve(i);

%% Electric smart pump
head_signal = (T_head(i-1) - 90)/(100 - 90);
block_signal = (T_block(i-1) - 85)/(105 - 85);
cool_signal = (T_cool(i-1) - 82)/(92 - 82);

head_signal = max(0, min(head_signal, 1));
block_signal = max(0, min(block_signal, 1));
cool_signal = max(0, min(cool_signal, 1));

temperature_demand = max([head_signal, cool_signal, valve(i)*block_signal]);

desired_pump = 0.22 + 0.08*load(i) + 0.30*temperature_demand;
desired_pump = max(0.22, min(desired_pump, 0.55));

pump(i) = pump(i-1) + dt*(desired_pump - pump(i-1))/tau_pump;
pump(i) = max(0.20, min(pump(i), 0.55));

%% Smart temperature targets
T_head_target = 87 + 12*load(i);
T_block_target = 80 + 25*load(i);
T_cool_target = 80 + 12*load(i);

%% Temperature response
T_head(i) = T_head(i-1) + dt*(T_head_target - T_head(i-1)) / tau_head;
T_block(i) = T_block(i-1) + dt*(T_block_target - T_block(i-1)) / tau_block;
T_cool(i) = T_cool(i-1) + dt*(T_cool_target - T_cool(i-1)) / tau_cool;

%% Radiator heat rejection

```

```

    Qrad(i) = valve(i) * (pump(i)/0.55) * 1200 * max(T_cool(i)-25,0);

end

rpm(1) = rpm(2);
load(1) = load(2);
Qrad(1) = Qrad(2);

%% Plot 1: Engine operation
figure;
subplot(2,1,1)
plot(t,rpm,'k','LineWidth',2.5);
grid on;
xlabel('Time (s)');
ylabel('RPM');
title('Engine Speed');

subplot(2,1,2)
plot(t,load,'k','LineWidth',2.5);
grid on;
xlabel('Time (s)');
ylabel('Load');
title('Engine Load');

%% Plot 2: Temperatures
figure;
plot(t,T_head,'b','LineWidth',2.5); hold on;
plot(t,T_block,'g','LineWidth',2.5);
plot(t,T_cool,'m','LineWidth',2.5);
grid on;
xlabel('Time (s)','FontSize',12);
ylabel('Temperature (C)','FontSize',12);
title('Smart Head-First Cooling System Temperatures','FontSize',13,'FontWeight','bold');
legend('Cylinder Head','Engine Block','Coolant','Location','southeast');
xlim([0 1400]);
ylim([30 110]);
set(gca,'FontSize',11);

%% Plot 3: Electronic 3-way control valve
figure;
plot(t, valve*100, 'b', 'LineWidth', 2.5); hold on;
plot(t, bypass*100, 'r--', 'LineWidth', 2.5);
grid on;
xlabel('Time (s)');
ylabel('Opening (%)');
title('Electronic 3-Way Control Valve Operation');
legend('Block/Radiator Path','Bypass Path','Location','best');
ylim([0 100]);
xlim([0 1400]);

%% Plot 4: Pump flow
figure;
plot(t,pump,'b','LineWidth',2.5);

```

```

grid on;
xlabel('Time (s)');
ylabel('Mass Flow Rate (kg/s)');
title('Electric Smart Pump Flow Rate');
legend('Electric Smart Pump','Location','best');
xlim([0 1400]);
ylim([0.20 0.60]);

%% Plot 5: Radiator heat rejection
figure;
plot(t,Qrad/1000,'b','LineWidth',2.5);
grid on;
xlabel('Time (s)');
ylabel('Radiator Heat Rejection (kW)');
title('Radiator Heat Rejection');
legend('Smart Cooling System','Location','best');
ylim([0 45]);
xlim([0 1400]);

%% Results
fprintf('\n===== SMART SYSTEM RESULTS =====\n');
fprintf('Final Head Temp      = %.2f C\n', T_head(end));
fprintf('Final Block Temp       = %.2f C\n', T_block(end));
fprintf('Final Coolant Temp      = %.2f C\n', T_cool(end));
fprintf('Final Pump Flow        = %.3f kg/s\n', pump(end));
fprintf('Final Valve Opening    = %.1f %%\n', valve(end)*100);
fprintf('Final Bypass Opening   = %.1f %%\n', bypass(end)*100);
fprintf('Final Radiator Heat    = %.2f kW\n', Qrad(end)/1000);
fprintf('===== \n');

disp('Conclusion:');
disp('The smart head-first system warms up efficiently using the bypass path. ');
disp('After warm-up, the electronic valve gradually opens the radiator path. ');
disp('The electric pump responds to temperature demand. ');
disp('The system keeps head, block, and coolant temperatures in the desired range. ');

```

APPENDIX B

MATLAB script for comparison :

```
clc; clear; close all;
```

```
%% SMART HEAD-FIRST COOLING SYSTEM VS CONVENTIONAL SYSTEM
```

```
%% Time
```

```
dt = 1;
```

```
t_end = 1400;
```

```
t = 0:dt:t_end;
```

```
n = length(t);
```

```
%% Initial temperatures
```

```
T_head_S = zeros(1,n);
```

```
T_block_S = zeros(1,n);
```

```
T_cool_S = zeros(1,n);
```

```
T_head_C = zeros(1,n);
```

```
T_block_C = zeros(1,n);
```

```
T_cool_C = zeros(1,n);
```

```
T_head_S(1) = 30;
```

```
T_block_S(1) = 30;
```

```
T_cool_S(1) = 30;
```

```
T_head_C(1) = 30;
```

```
T_block_C(1) = 30;
```

```
T_cool_C(1) = 30;
```

```
%% Variables
```

```
rpm = zeros(1,n);
```

```
load = zeros(1,n);
```

```
pump_S = zeros(1,n);
```

```
valve_S = zeros(1,n);
```

```
bypass_S = zeros(1,n);
```

```
Qrad_S = zeros(1,n);
```

```
pump_C = zeros(1,n);
```

```
valve_C = zeros(1,n);
```

```
Qrad_C = zeros(1,n);
```

```
pump_S(1) = 0.22;
```

```
valve_S(1) = 0.00;
```

```

bypass_S(1) = 1.00;

pump_C(1) = 0.20;
valve_C(1) = 0.10;

%% Response constants
tau_head_S = 80;
tau_block_S = 145;
tau_cool_S = 105;
tau_pump_S = 75;

tau_head_C = 115;
tau_block_C = 185;
tau_cool_C = 145;

%% Simulation
for i = 2:n

    %% Engine operation: light -> medium -> high
    load(i) = 0.35 ...
        + 0.25/(1 + exp(-(t(i)-350)/35)) ...
        + 0.20/(1 + exp(-(t(i)-700)/45));

    rpm(i) = 1500 ...
        + 1200/(1 + exp(-(t(i)-350)/35)) ...
        + 1300/(1 + exp(-(t(i)-700)/45));

    %% SMART HEAD-FIRST SYSTEM

    %% Electronic 3-way control valve
    if t(i) < 250
        desired_valve_S = 0.00;
    elseif t(i) < 650
        desired_valve_S = 0.35*(t(i)-250)/(650-250);
    elseif t(i) < 850
        desired_valve_S = 0.35 + 0.10*(t(i)-650)/(850-650);
    elseif t(i) < 1050
        desired_valve_S = 0.45 + 0.37*(t(i)-850)/(1050-850);
    else
        desired_valve_S = 0.82;
    end

    valve_S(i) = valve_S(i-1) + dt*(desired_valve_S - valve_S(i-1))/35;
    valve_S(i) = max(0.00, min(valve_S(i), 0.85));
    bypass_S(i) = 1 - valve_S(i);

    %% Smart electric pump
    head_signal = (T_head_S(i-1) - 90)/(100 - 90);
    block_signal = (T_block_S(i-1) - 85)/(105 - 85);
    cool_signal = (T_cool_S(i-1) - 82)/(92 - 82);

    head_signal = max(0, min(head_signal, 1));
    block_signal = max(0, min(block_signal, 1));

```

```

cool_signal = max(0, min(cool_signal, 1));

temperature_demand = max([head_signal, cool_signal, valve_S(i)*block_signal]);

desired_pump_S = 0.22 + 0.08*load(i) + 0.30*temperature_demand;
desired_pump_S = max(0.22, min(desired_pump_S, 0.55));

pump_S(i) = pump_S(i-1) + dt*(desired_pump_S - pump_S(i-1))/tau_pump_S;
pump_S(i) = max(0.20, min(pump_S(i), 0.55));

%% Smart temperature targets
T_head_target_S = 87 + 12*load(i);
T_block_target_S = 80 + 25*load(i);
T_cool_target_S = 80 + 12*load(i);

%% Smart temperature response
T_head_S(i) = T_head_S(i-1) + dt*(T_head_target_S - T_head_S(i-1)) / tau_head_S;
T_block_S(i) = T_block_S(i-1) + dt*(T_block_target_S - T_block_S(i-1)) / tau_block_S;
T_cool_S(i) = T_cool_S(i-1) + dt*(T_cool_target_S - T_cool_S(i-1)) / tau_cool_S;

Qrad_S(i) = valve_S(i) * (pump_S(i)/0.55) * 1200 * max(T_cool_S(i)-25,0);

%% CONVENTIONAL COOLING SYSTEM

%% Conventional thermostat
if T_cool_C(i-1) < 82
    desired_valve_C = 0.10;
elseif T_cool_C(i-1) > 98
    desired_valve_C = 1.00;
else
    desired_valve_C = 0.10 + 0.90*(T_cool_C(i-1)-82)/(98-82);
end

valve_C(i) = valve_C(i-1) + dt*(desired_valve_C - valve_C(i-1))/90;
valve_C(i) = max(0.10, min(valve_C(i), 1.00));

%% Mechanical pump depends mainly on RPM
rpm_signal = (rpm(i)-1500)/(4000-1500);
rpm_signal = max(0, min(rpm_signal, 1));

pump_C(i) = 0.18 + 0.26*rpm_signal;
pump_C(i) = max(0.18, min(pump_C(i), 0.44));

%% Conventional temperature targets
T_head_target_C = 92 + 16*load(i);
T_block_target_C = 84 + 27*load(i);
T_cool_target_C = 84 + 14*load(i);

if t(i) < 350
    T_head_target_C = T_head_target_C - 10;
    T_block_target_C = T_block_target_C - 8;
    T_cool_target_C = T_cool_target_C - 9;

```

```

    tau_head_C_now = 170;
    tau_block_C_now = 240;
    tau_cool_C_now = 200;
else
    tau_head_C_now = tau_head_C;
    tau_block_C_now = tau_block_C;
    tau_cool_C_now = tau_cool_C;
end

%% Conventional temperature response
T_head_C(i) = T_head_C(i-1) + dt*(T_head_target_C - T_head_C(i-1)) /
tau_head_C_now;
T_block_C(i) = T_block_C(i-1) + dt*(T_block_target_C - T_block_C(i-1)) /
tau_block_C_now;
T_cool_C(i) = T_cool_C(i-1) + dt*(T_cool_target_C - T_cool_C(i-1)) /
tau_cool_C_now;

Qrad_C(i) = valve_C(i) * (pump_C(i)/0.50) * 1200 * max(T_cool_C(i)-25,0);

end

rpm(1) = rpm(2);
load(1) = load(2);
Qrad_S(1) = Qrad_S(2);
Qrad_C(1) = Qrad_C(2);

%% Warm-up time analysis
targetWarmup = 85;

idxWarmS = find(T_head_S >= targetWarmup,1);
idxWarmC = find(T_head_C >= targetWarmup,1);

warmTimeS = t(idxWarmS);
warmTimeC = t(idxWarmC);
warmImprovement = ((warmTimeC - warmTimeS)/warmTimeC)*100;

%% Plot 1: Engine operation
figure;
subplot(2,1,1)
plot(t,rpm,'k','LineWidth',2.5);
grid on;
xlabel('Time (s)');
ylabel('RPM');
title('Engine Speed');

subplot(2,1,2)
plot(t,load,'k','LineWidth',2.5);
grid on;
xlabel('Time (s)');
ylabel('Load');
title('Engine Load');

%% Plot 2A: Cylinder head temperature comparison

```

```

figure;
plot(t,T_head_S,'b','LineWidth',2.5); hold on;
plot(t,T_head_C,'r--','LineWidth',2.5);
yline(targetWarmup,'k:','85 C Warm-up Target','LineWidth',1.2);
xline(warmTimeS,'b:','Smart reaches 85 C','LineWidth',1.2);
xline(warmTimeC,'r:','Conventional reaches 85 C','LineWidth',1.2);
grid on;
xlabel('Time (s)','FontSize',12);
ylabel('Head Temperature (C)','FontSize',12);
title('Cylinder Head Temperature Comparison','FontSize',13,'FontWeight','bold');
legend('Smart Head-First','Conventional','Location','southeast');
xlim([0 1400]);
ylim([30 110]);
set(gca,'FontSize',11);

```

```

%% Plot 2D: Warm-up time comparison

```

```

figure;
bar([warmTimeS warmTimeC]);
grid on;
ylabel('Warm-up Time to 85 C (s)','FontSize',12);
title('Warm-up Time Comparison','FontSize',13,'FontWeight','bold');
set(gca,'XTickLabel',{'Smart Head-First','Conventional'});
set(gca,'FontSize',11);

```

```

%% Plot 2B: Engine block temperature comparison

```

```

figure;
plot(t,T_block_S,'b','LineWidth',2.5); hold on;
plot(t,T_block_C,'r--','LineWidth',2.5);
grid on;
xlabel('Time (s)','FontSize',12);
ylabel('Block Temperature (C)','FontSize',12);
title('Engine Block Temperature Comparison','FontSize',13,'FontWeight','bold');
legend('Smart Head-First','Conventional','Location','southeast');
xlim([0 1400]);
ylim([30 110]);
set(gca,'FontSize',11);

```

```

%% Plot 2C: Coolant temperature comparison

```

```

figure;
plot(t,T_cool_S,'b','LineWidth',2.5); hold on;
plot(t,T_cool_C,'r--','LineWidth',2.5);
grid on;
xlabel('Time (s)','FontSize',12);
ylabel('Coolant Temperature (C)','FontSize',12);
title('Coolant Temperature Comparison','FontSize',13,'FontWeight','bold');
legend('Smart Head-First','Conventional','Location','southeast');
xlim([0 1400]);
ylim([20 105]);
set(gca,'FontSize',11);

```

```

%% Plot 3: Smart valve operation

```

```

figure;
plot(t,valve_S*100,'b','LineWidth',2.5); hold on;

```

```

plot(t,bypass_S*100,'r--','LineWidth',2.5);
grid on;
xlabel('Time (s)');
ylabel('Opening (%)');
title('Electronic 3-Way Control Valve Operation');
legend('Block/Radiator Path','Bypass Path','Location','best');
ylim([0 100]);
xlim([0 1400]);

%% Plot 4: Pump comparison
figure;
plot(t,pump_S,'b','LineWidth',2.5); hold on;
plot(t,pump_C,'r--','LineWidth',2.5);
grid on;
xlabel('Time (s)');
ylabel('Mass Flow Rate (kg/s)');
title('Pump Flow Rate Comparison');
legend('Electric Smart Pump','Mechanical Pump','Location','best');

%% Plot 5: Radiator heat rejection comparison
figure;
plot(t,Qrad_S/1000,'b','LineWidth',2.5); hold on;
plot(t,Qrad_C/1000,'r--','LineWidth',2.5);
grid on;
xlabel('Time (s)');
ylabel('Radiator Heat Rejection (kW)');
title('Radiator Heat Rejection Comparison');
legend('Smart Cooling System','Conventional Cooling System','Location','best');
ylim([0 45]);
xlim([0 1400]);

%% Results
fprintf('\n===== FINAL COMPARISON RESULTS\n');

fprintf('\n--- Smart System ---\n');
fprintf('Final Head Temp   = %.2f C\n', T_head_S(end));
fprintf('Final Block Temp   = %.2f C\n', T_block_S(end));
fprintf('Final Coolant Temp = %.2f C\n', T_cool_S(end));
fprintf('Final Pump Flow    = %.3f kg/s\n', pump_S(end));
fprintf('Final Valve Opening= %.1f %%\n', valve_S(end)*100);

fprintf('\n--- Conventional System ---\n');
fprintf('Final Head Temp   = %.2f C\n', T_head_C(end));
fprintf('Final Block Temp   = %.2f C\n', T_block_C(end));
fprintf('Final Coolant Temp = %.2f C\n', T_cool_C(end));
fprintf('Final Pump Flow    = %.3f kg/s\n', pump_C(end));
fprintf('Final Thermostat   = %.1f %%\n', valve_C(end)*100);

fprintf('\n--- Improvement ---\n');
fprintf('Smart Warm-up Time to 85 C   = %.1f s\n', warmTimeS);
fprintf('Conventional Warm-up Time to 85 C = %.1f s\n', warmTimeC);
fprintf('Warm-up Improvement          = %.1f %%\n', warmImprovement);

```

```

fprintf('Head Temperature Reduction   = %.2f C\n', T_head_C(end)-T_head_S(end));
fprintf('Block Temperature Reduction  = %.2f C\n', T_block_C(end)-T_block_S(end));
fprintf('Coolant Temperature Reduction = %.2f C\n', T_cool_C(end)-T_cool_S(end));
fprintf('=====\\n')
;

disp('Conclusion:');
disp('The smart head-first system warms up faster and then keeps the cylinder head cooler.');
```

VITA

MAJD NASER ABDALLAH MOFLEH is a senior Automotive Engineering student at Istanbul Okan University, Türkiye. Throughout his academic journey, he has developed a strong interest in internal combustion engines, automotive thermal management systems, heat transfer, fluid flow analysis, and advanced vehicle technologies. His studies focused on combining theoretical engineering knowledge with practical design and simulation approaches to solve real automotive engineering problems.

During his undergraduate education, he participated in various engineering projects and technical studies related to engine systems, cooling technologies, thermal analysis, and vehicle performance optimization. He gained experience in engineering design methodologies, analytical calculations, CAD modeling, and numerical simulation tools used in modern automotive engineering applications.

His graduation project, titled “Design and Thermal Analysis of an Improved Cooling System for a Four-Cylinder Gasoline Engine,” focuses on the development of a smart head-first cooling system integrating an electric water pump, electronically controlled bypass valve, and split-cooling architecture. The project includes thermal analysis, coolant flow control, CAD-based system modeling, and MATLAB script-based thermal simulations to evaluate engine temperature behavior under different operating conditions.

His technical interests include intelligent cooling systems, engine efficiency improvement, thermal management optimization, automotive simulation, and advanced powertrain technologies. He is particularly interested in the integration of electronic control systems with mechanical and thermal engineering applications to improve engine reliability, fuel efficiency, and overall vehicle performance.

MAJD NASER ABDALLAH MOFLEH aims to continue developing his technical and professional skills in the automotive engineering field and pursue a career focused on modern engine technologies, thermal systems, and advanced automotive innovation.